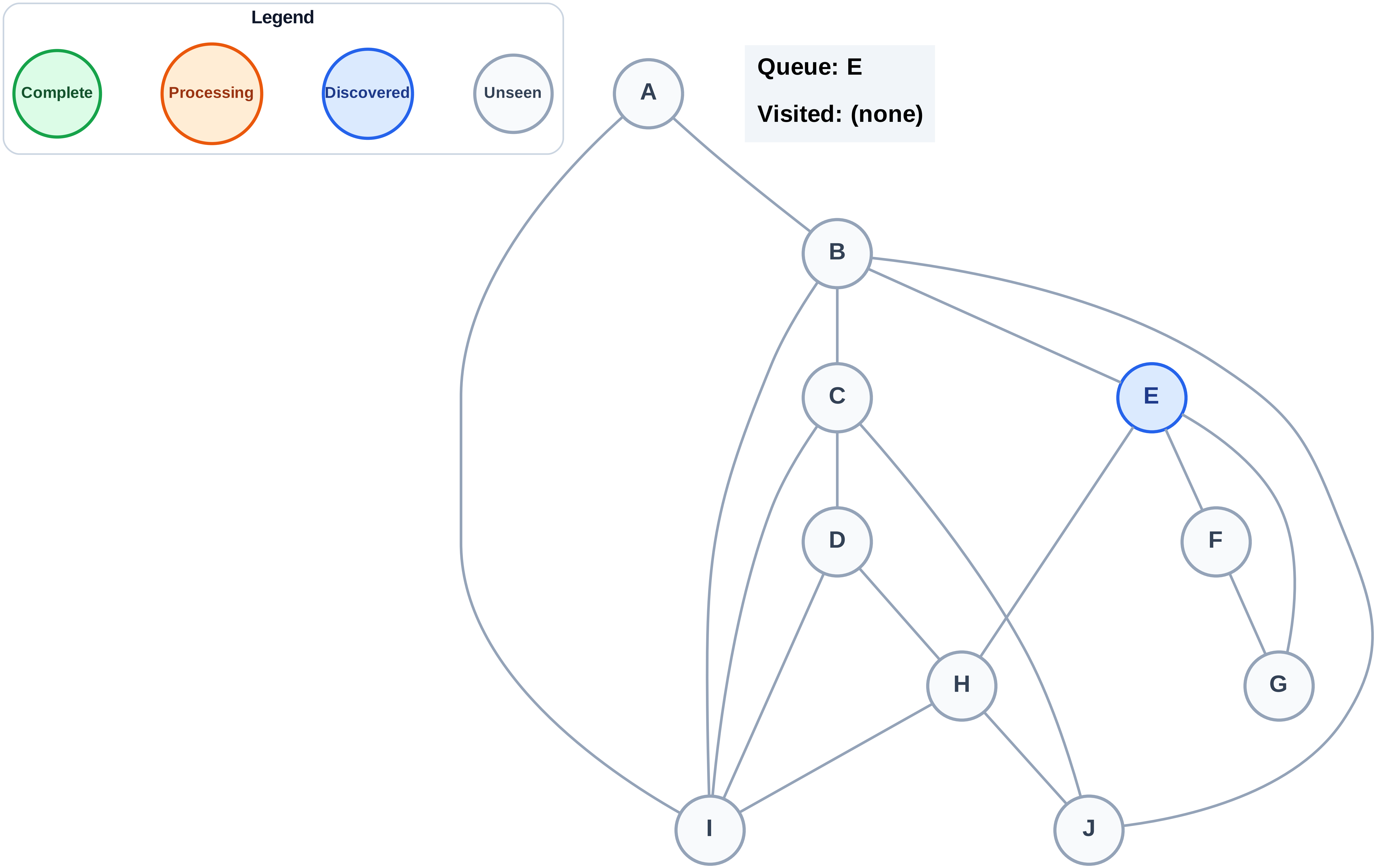
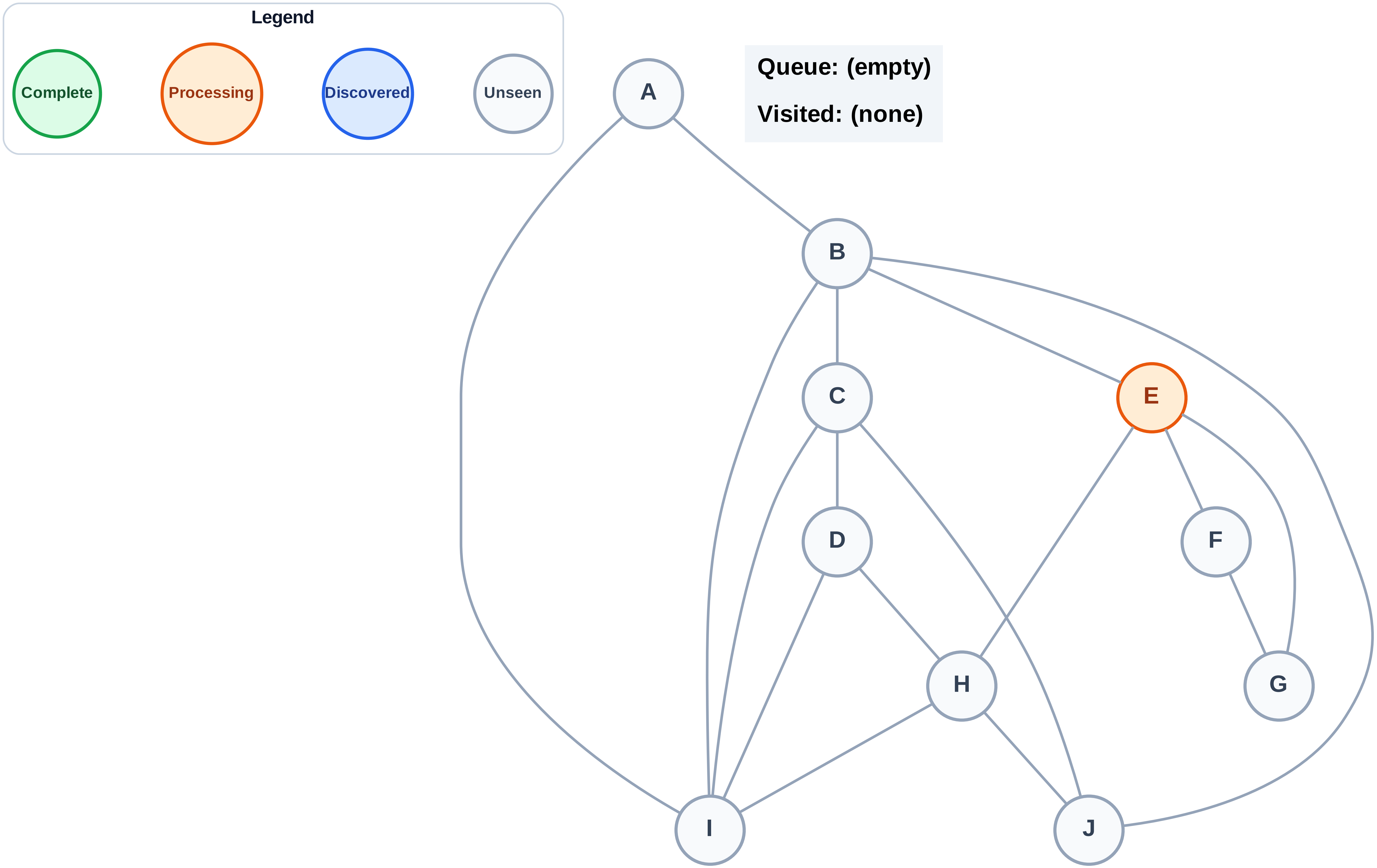
BFS Traversal

Step 1: Start at E



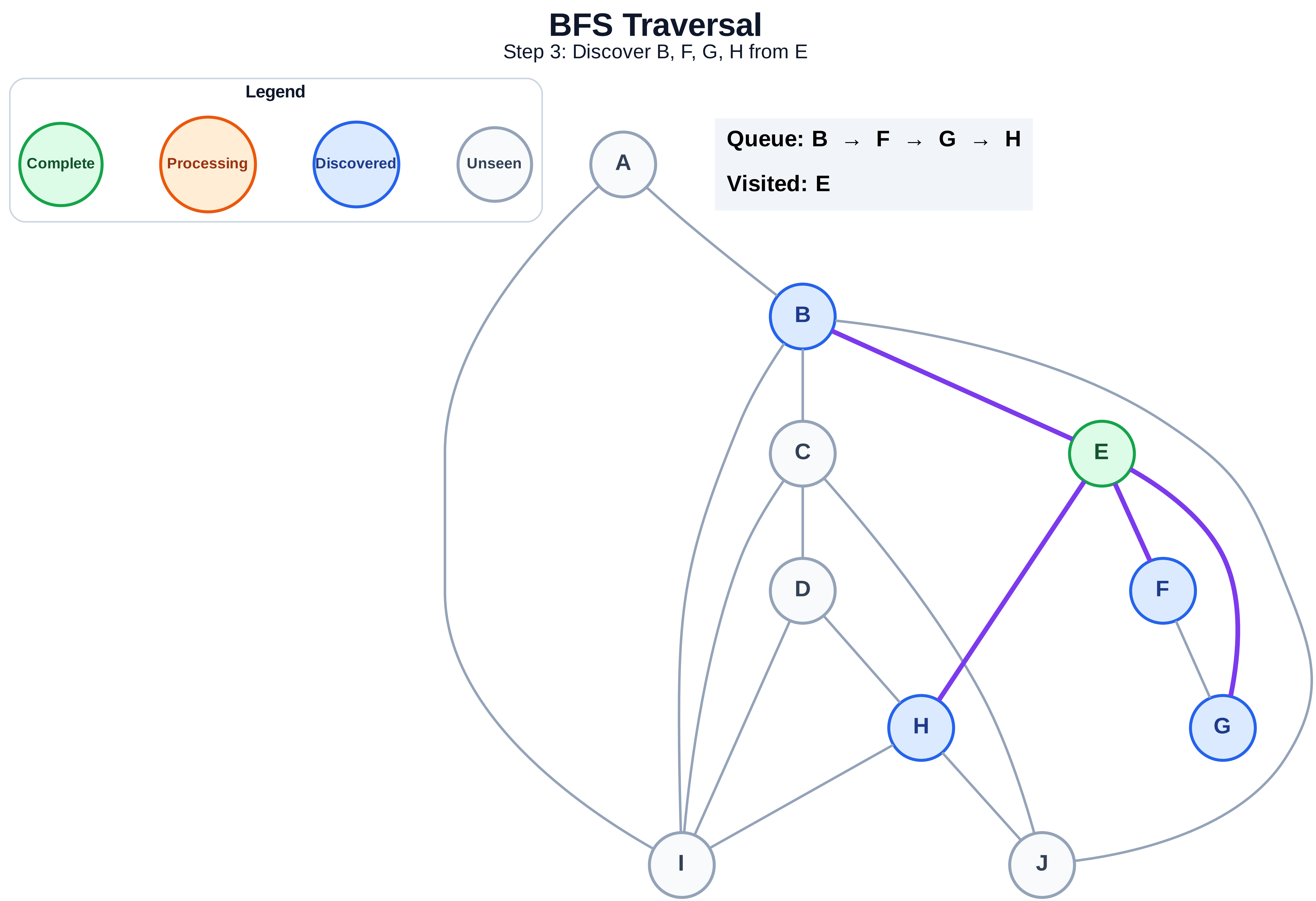
BFS Traversal

Step 2: Process node E



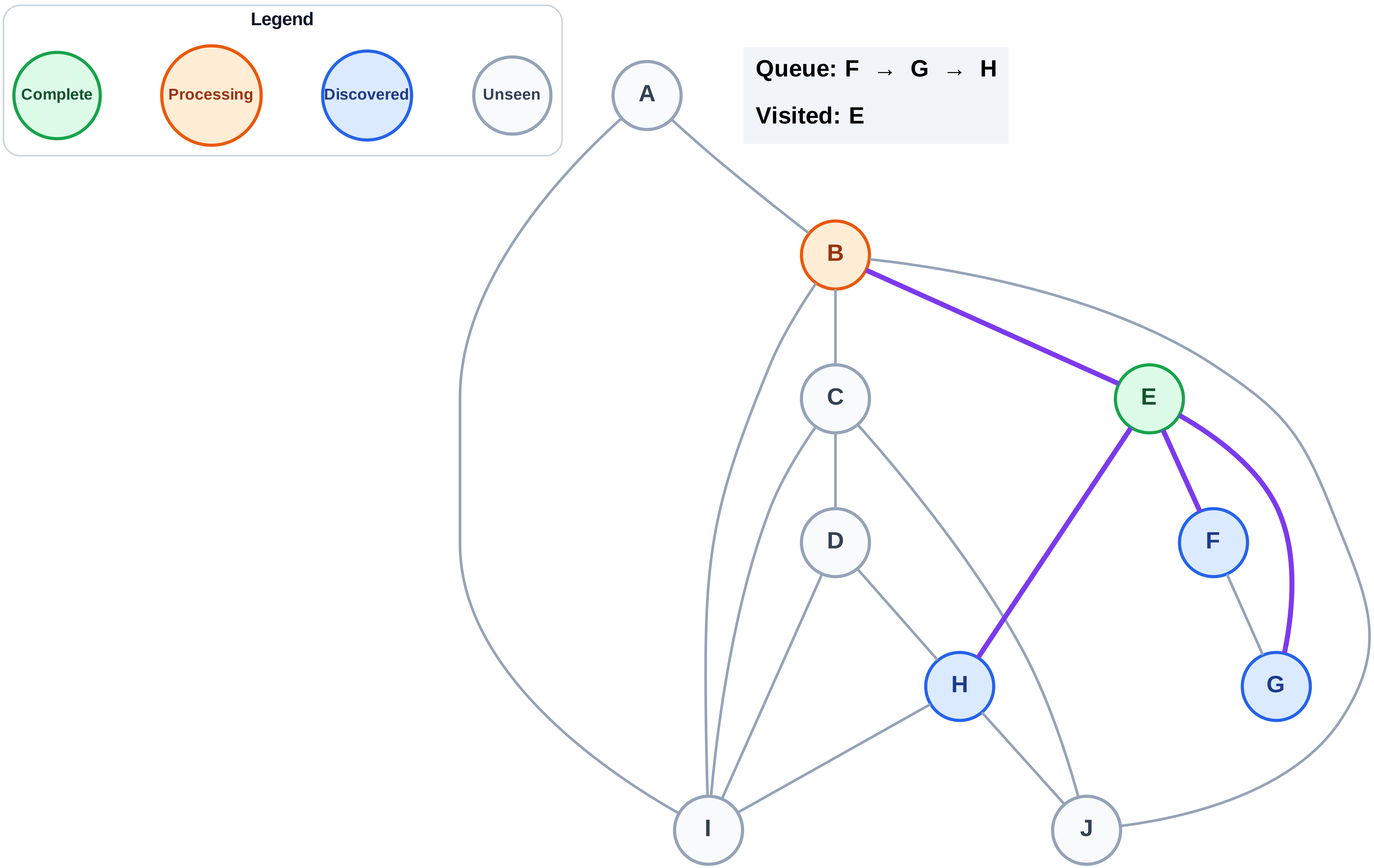
Queue: (empty)

Visited: (none)


BFS Traversal
Step 3: Discover B, F, G, H from E
Legend
Complete
Processing
Discovered
Unseen
Queue: B → F → G → H
Visited: E
A
B
C
D
E
F
G
H
I
J

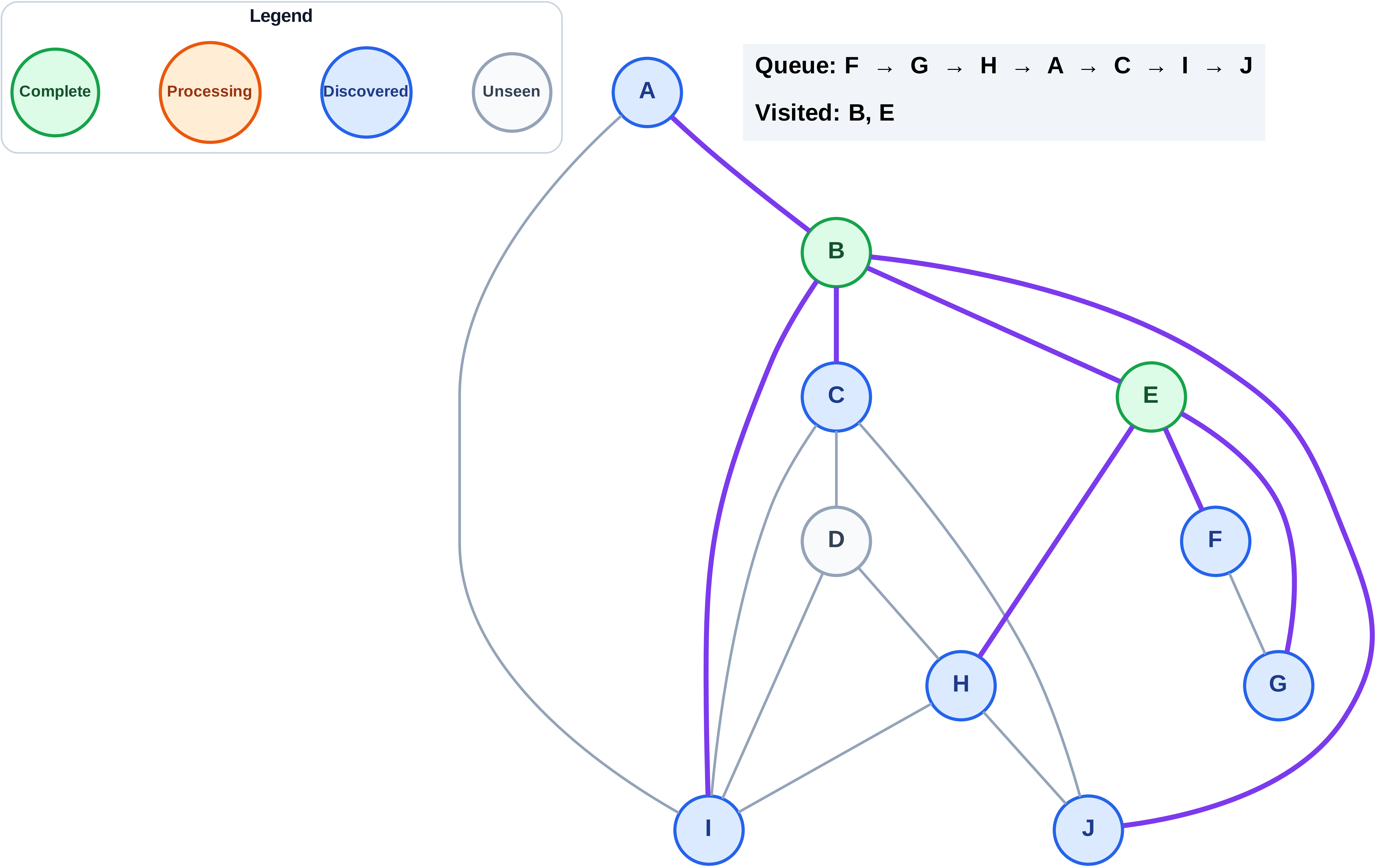
BFS Traversal

Step 4: Process node B



BFS Traversal

Step 5: Discover A, C, I, J from B

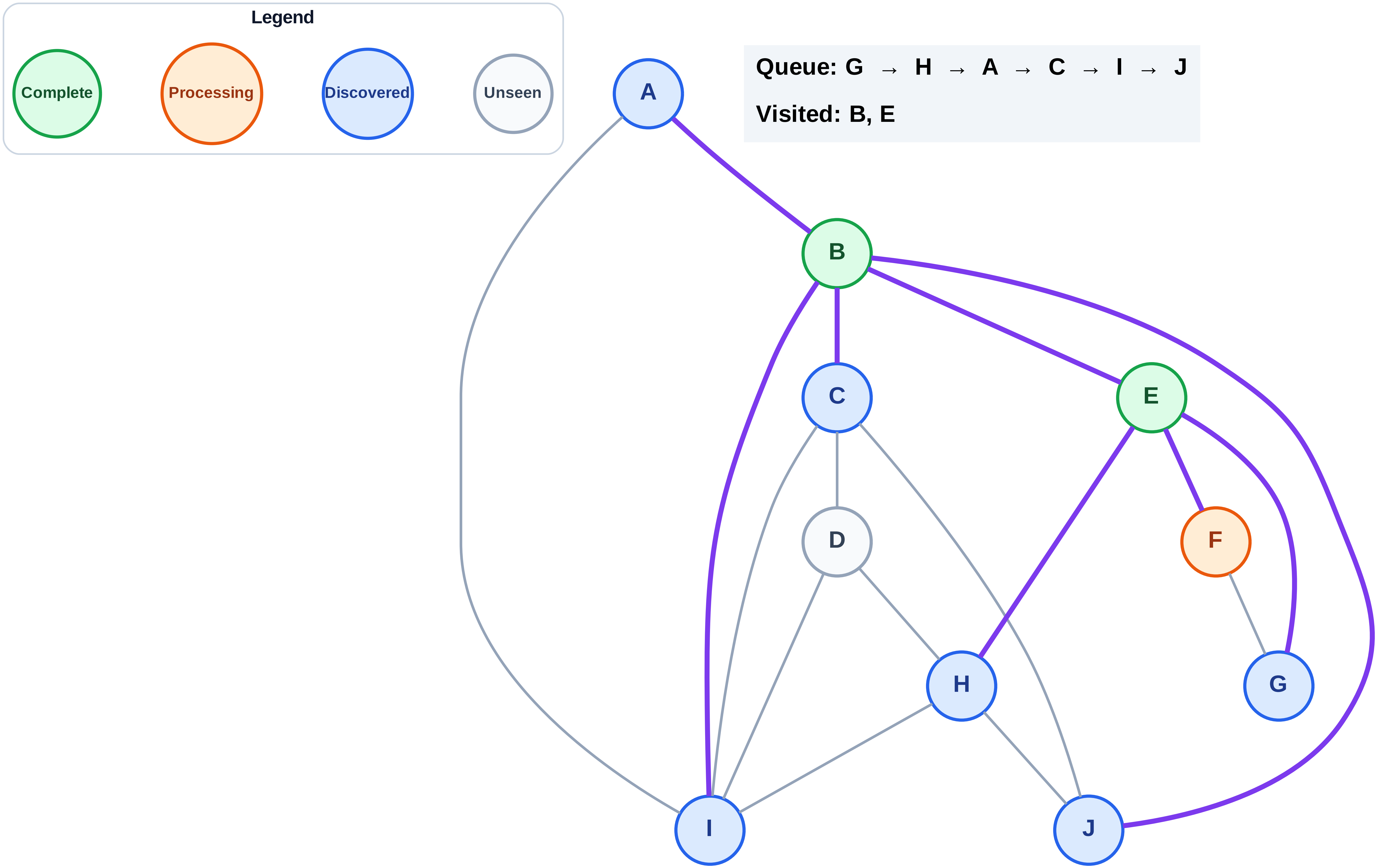


Queue: F → G → H → A → C → I → J

Visited: B, E

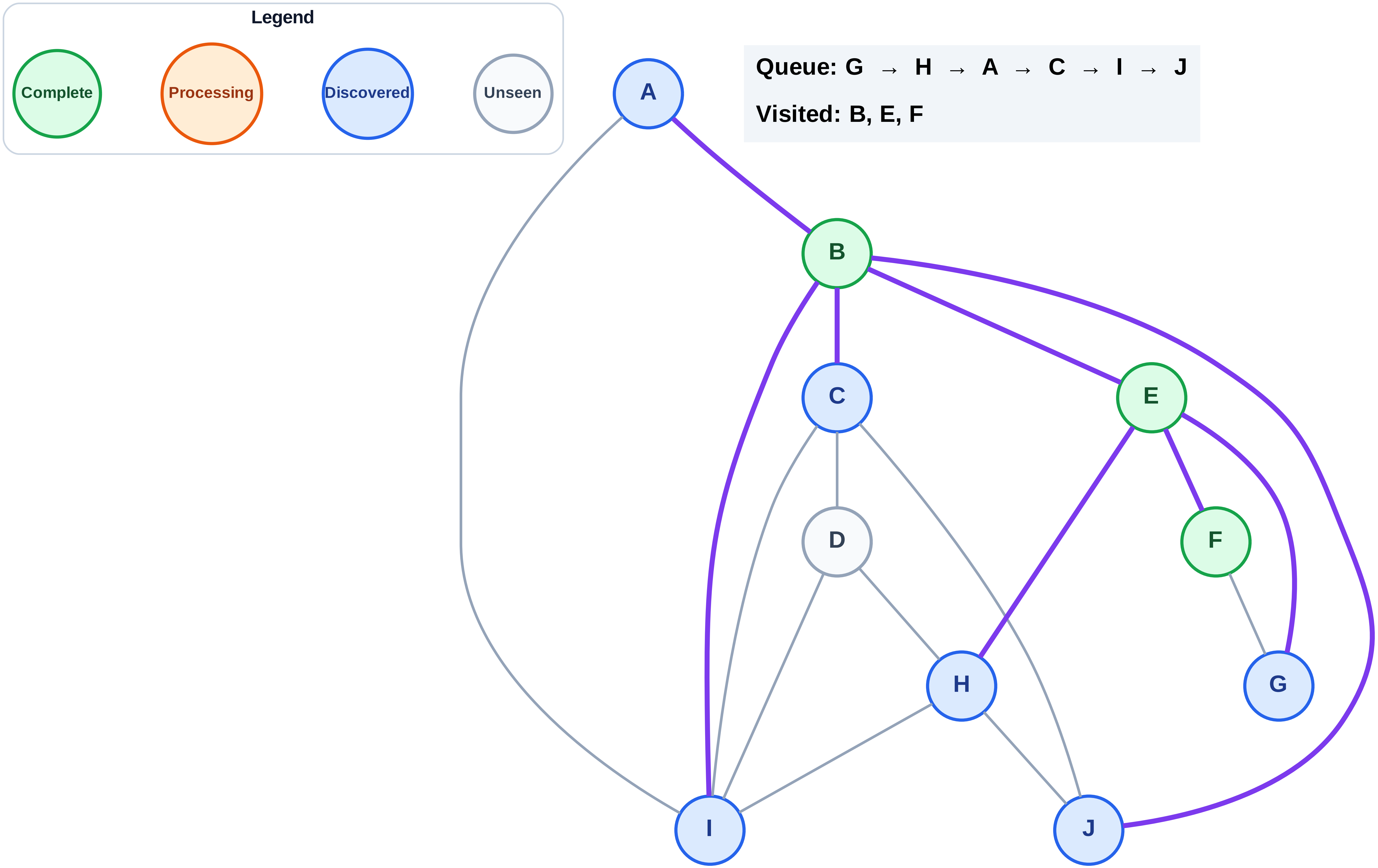
BFS Traversal

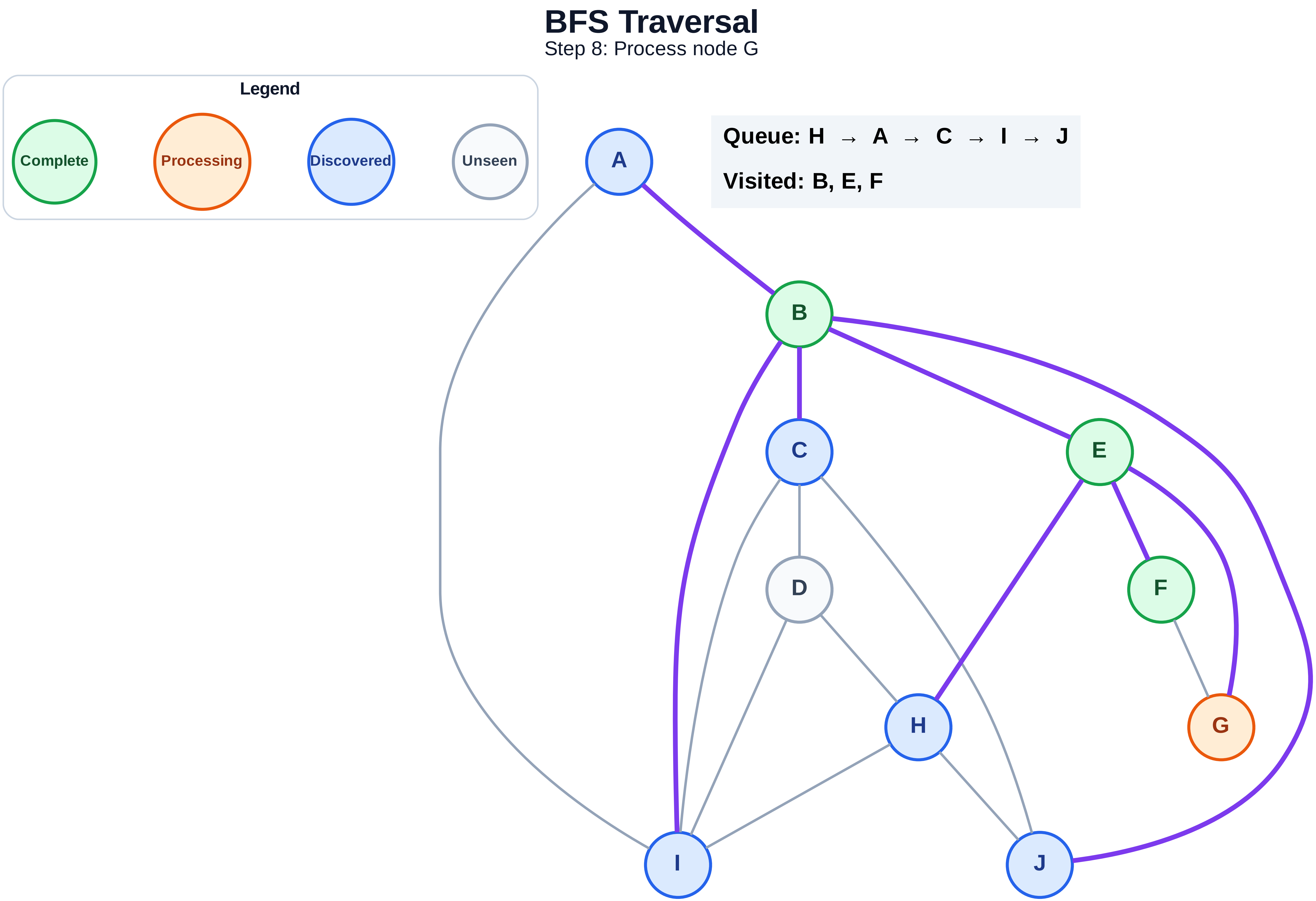
Step 6: Process node F

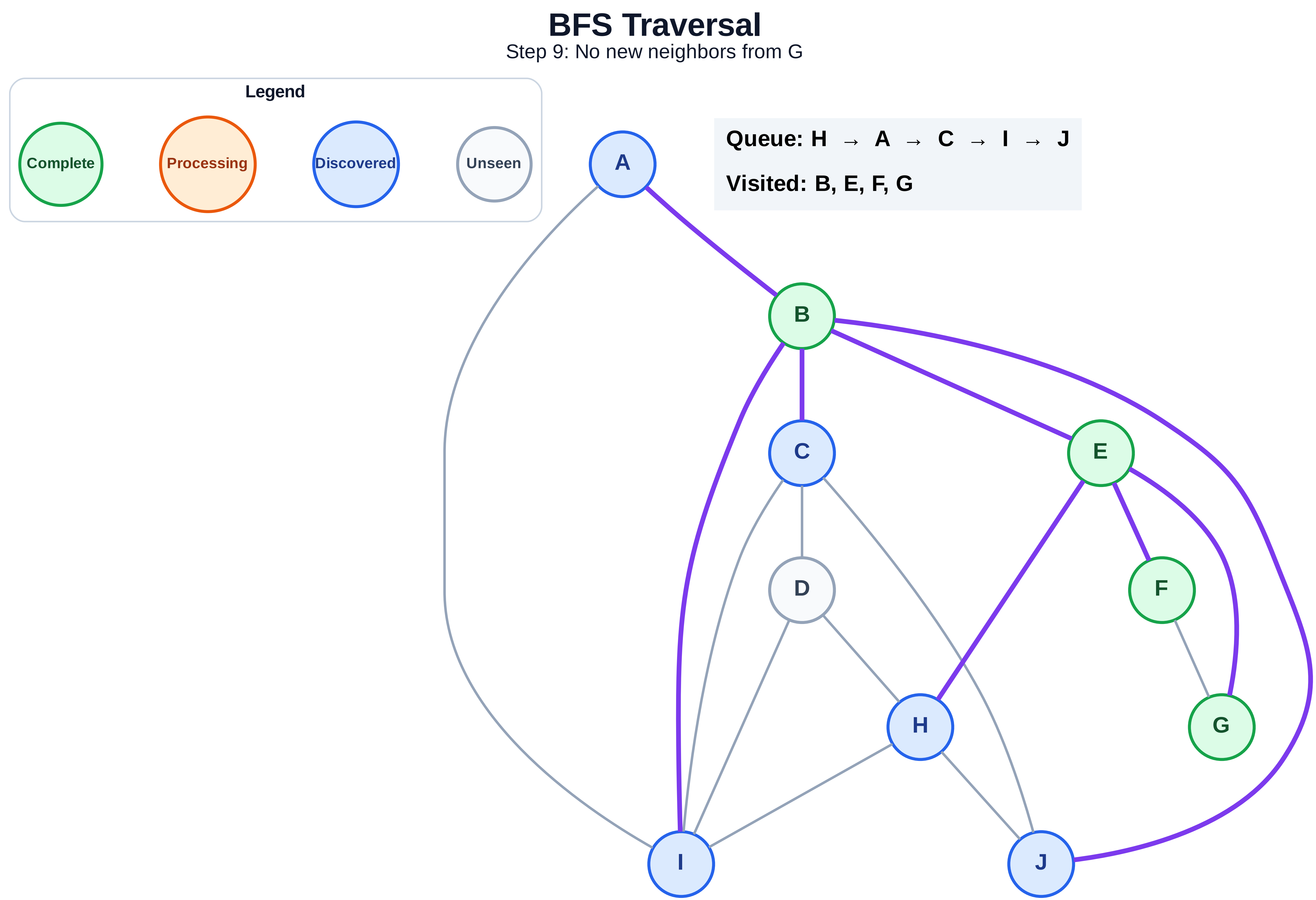


BFS Traversal

Step 7: No new neighbors from F

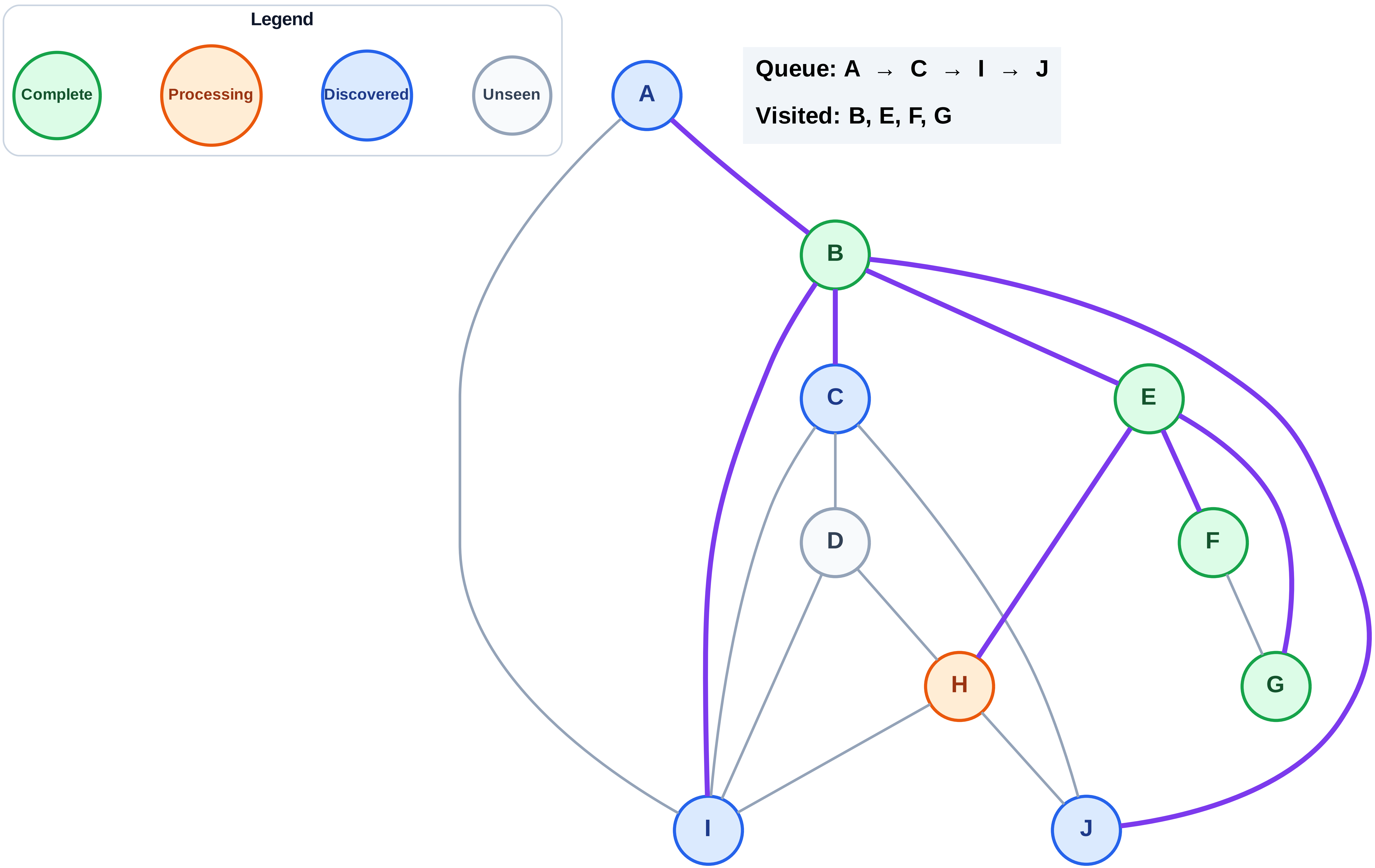



BFS Traversal
Step 8: Process node G
Legend
Complete
Processing
Discovered
Unseen
Queue: H → A → C → I → J
Visited: B, E, F
A
B
C
D
E
F
G
H
I
J


BFS Traversal
Step 9: No new neighbors from G
Legend
Complete
Processing
Discovered
Unseen
Queue: H → A → C → I → J
Visited: B, E, F, G
A
B
C
D
E
F
G
H
I
J

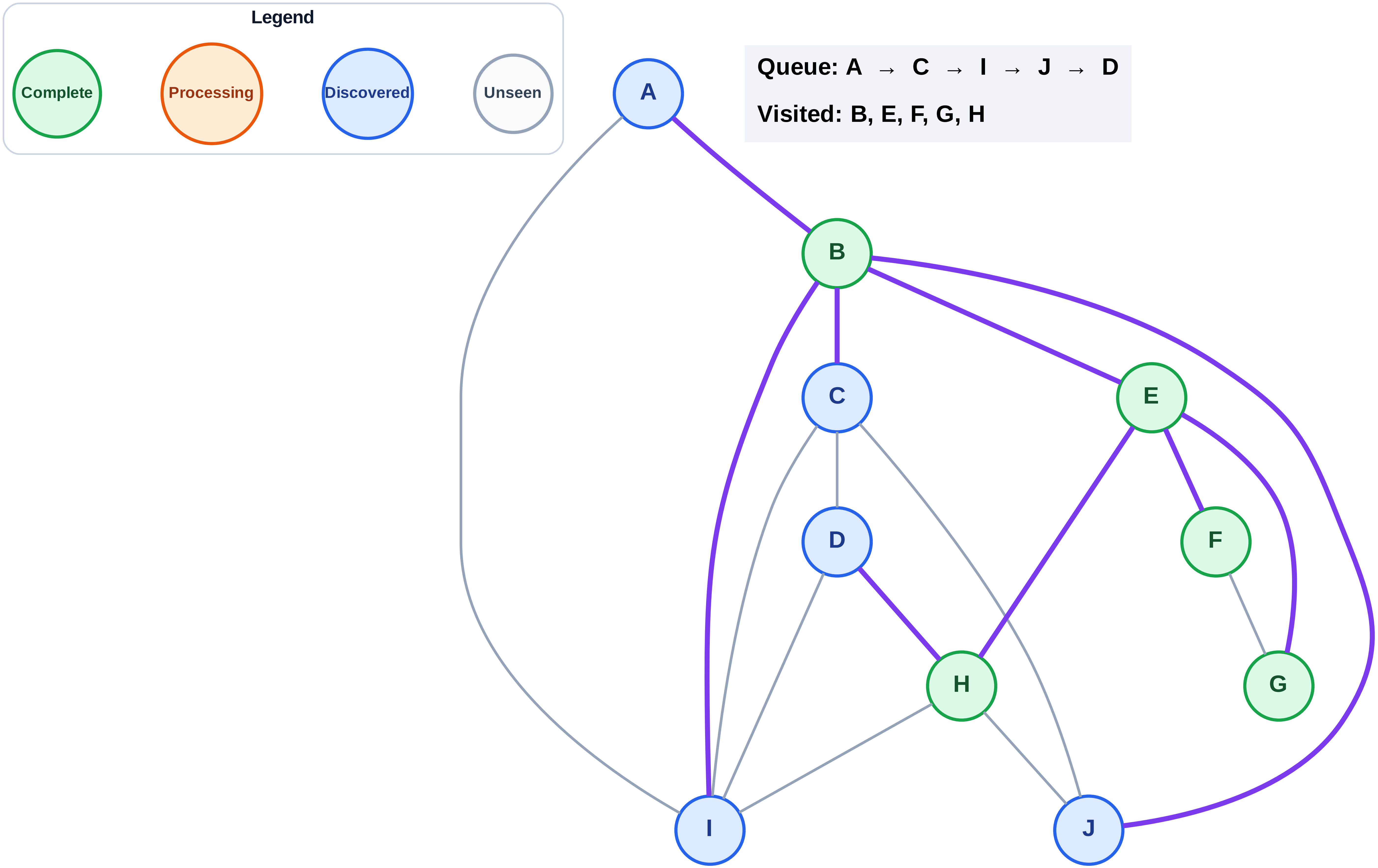
BFS Traversal

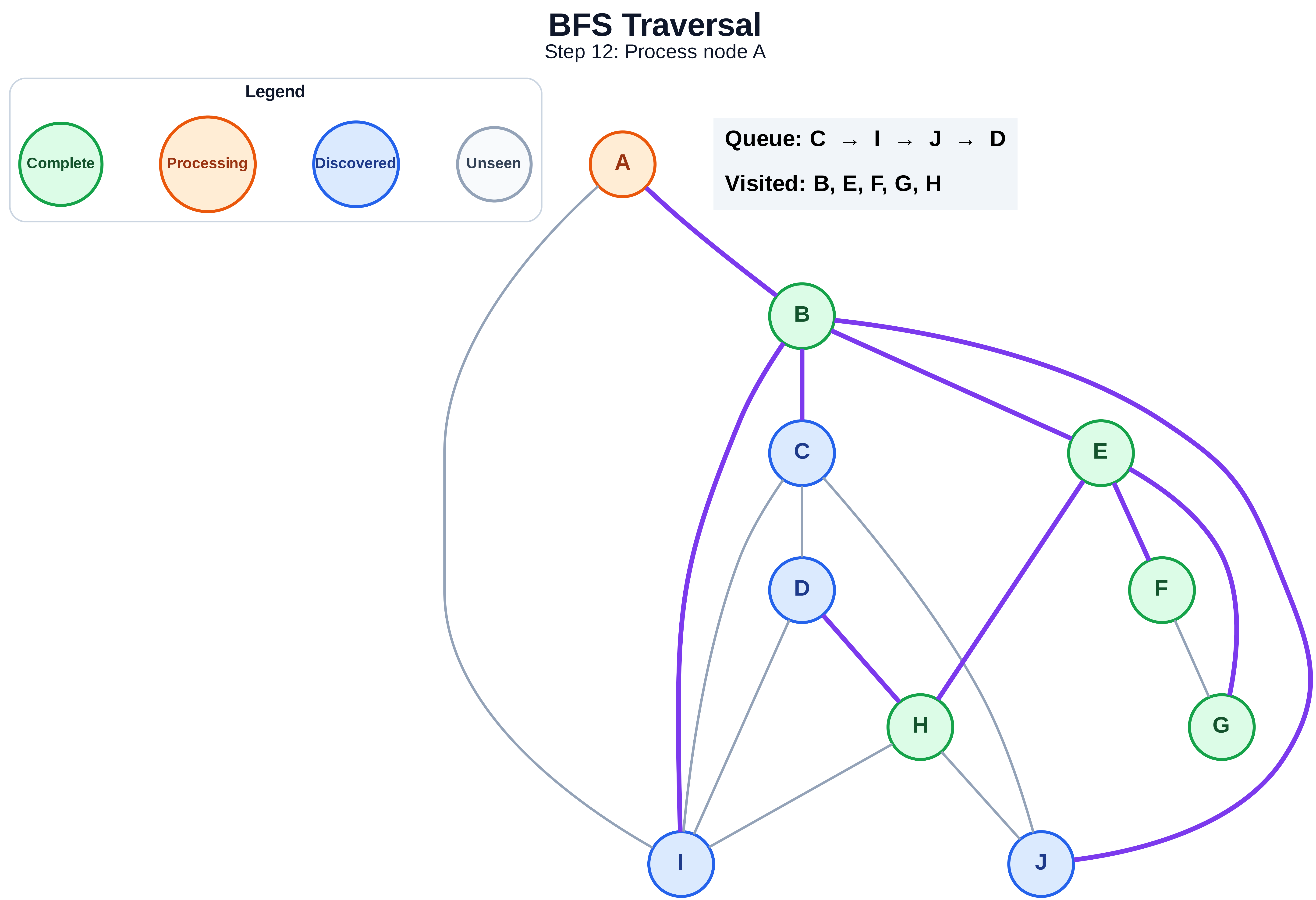
Step 10: Process node H



BFS Traversal

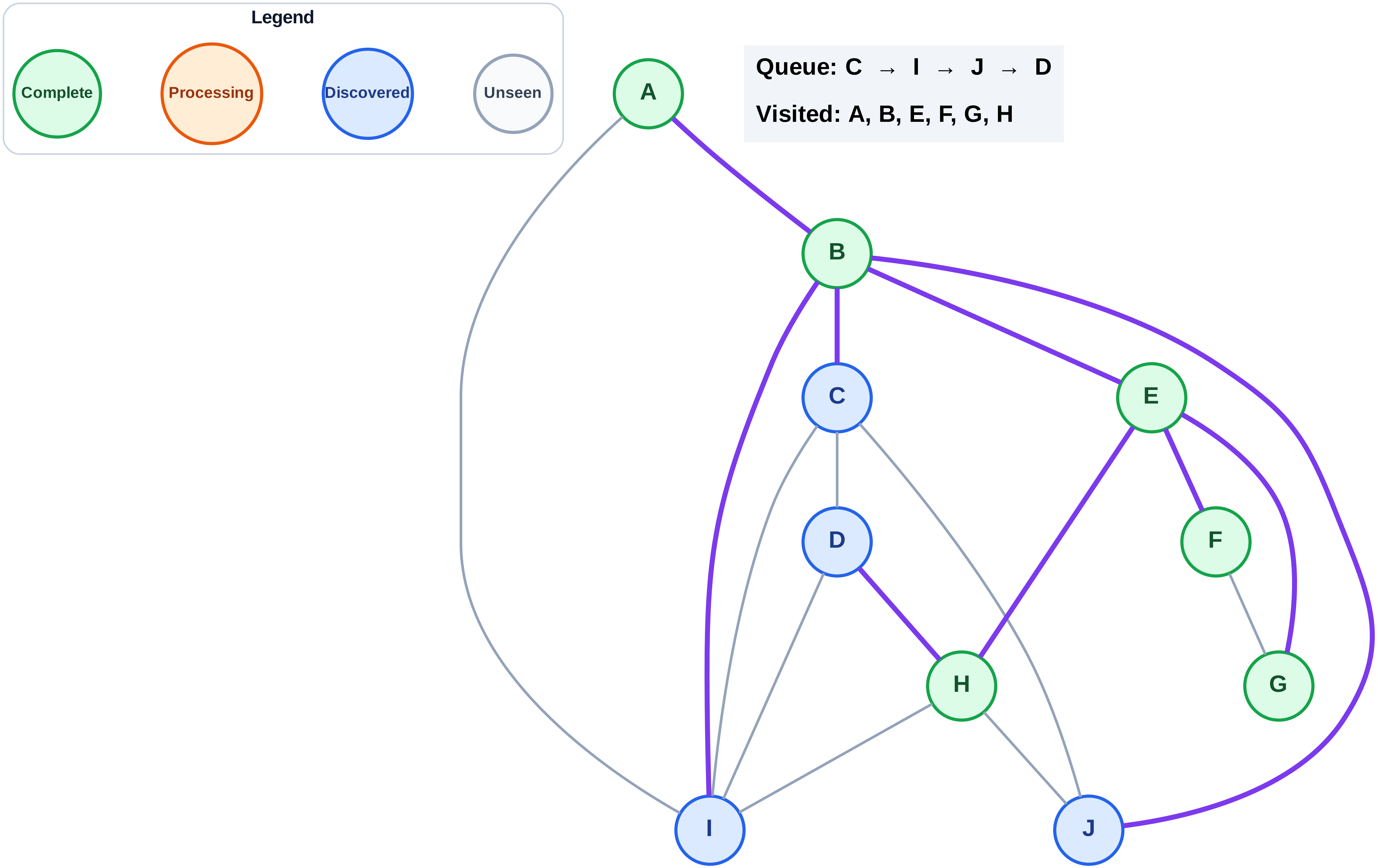
Step 11: Discover D from H




BFS Traversal
Step 12: Process node A
Legend
Complete
Processing
Discovered
Unseen
Queue: C → I → J → D
Visited: B, E, F, G, H
A
B
C
E
D
F
H
G
I
J

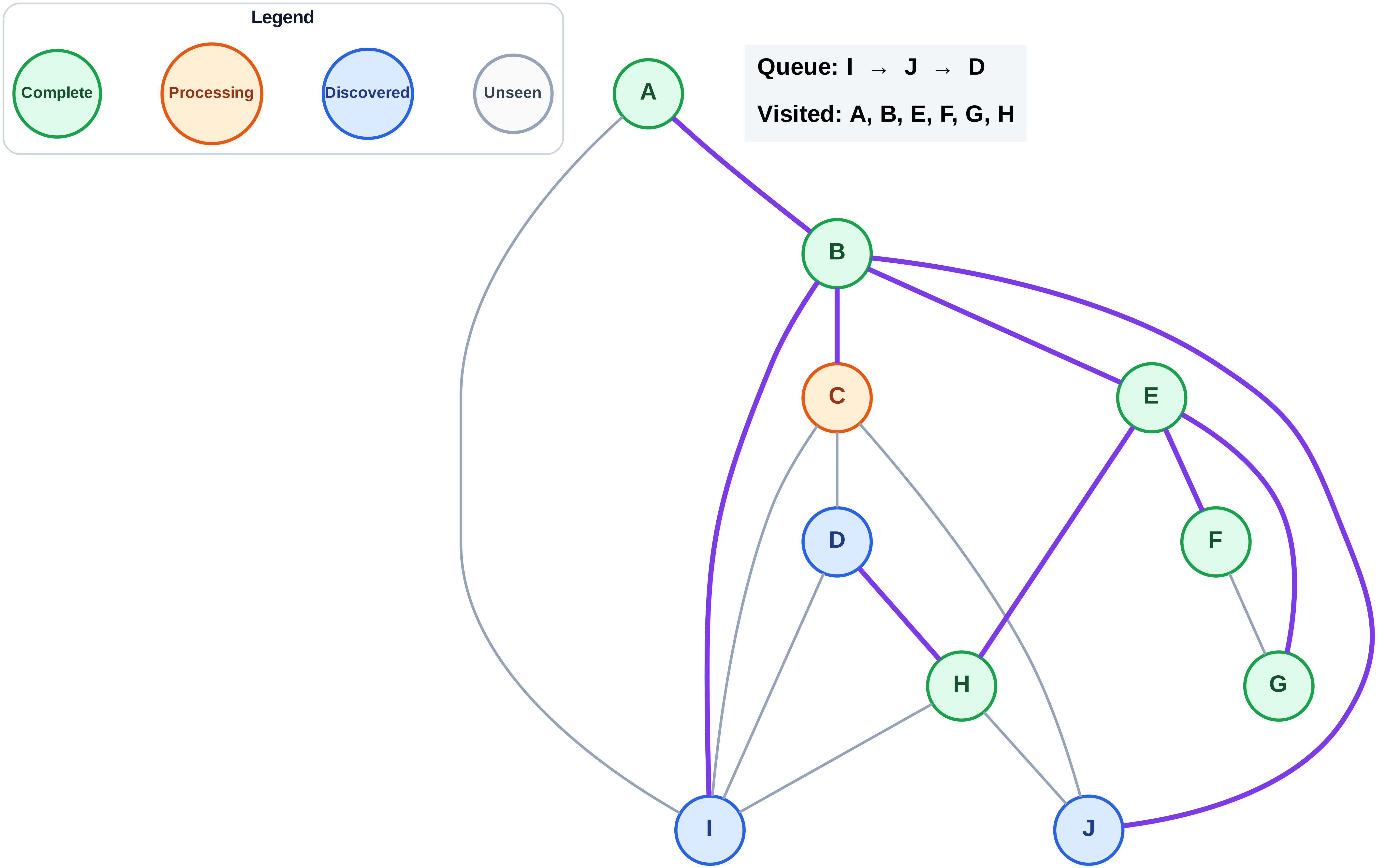
BFS Traversal

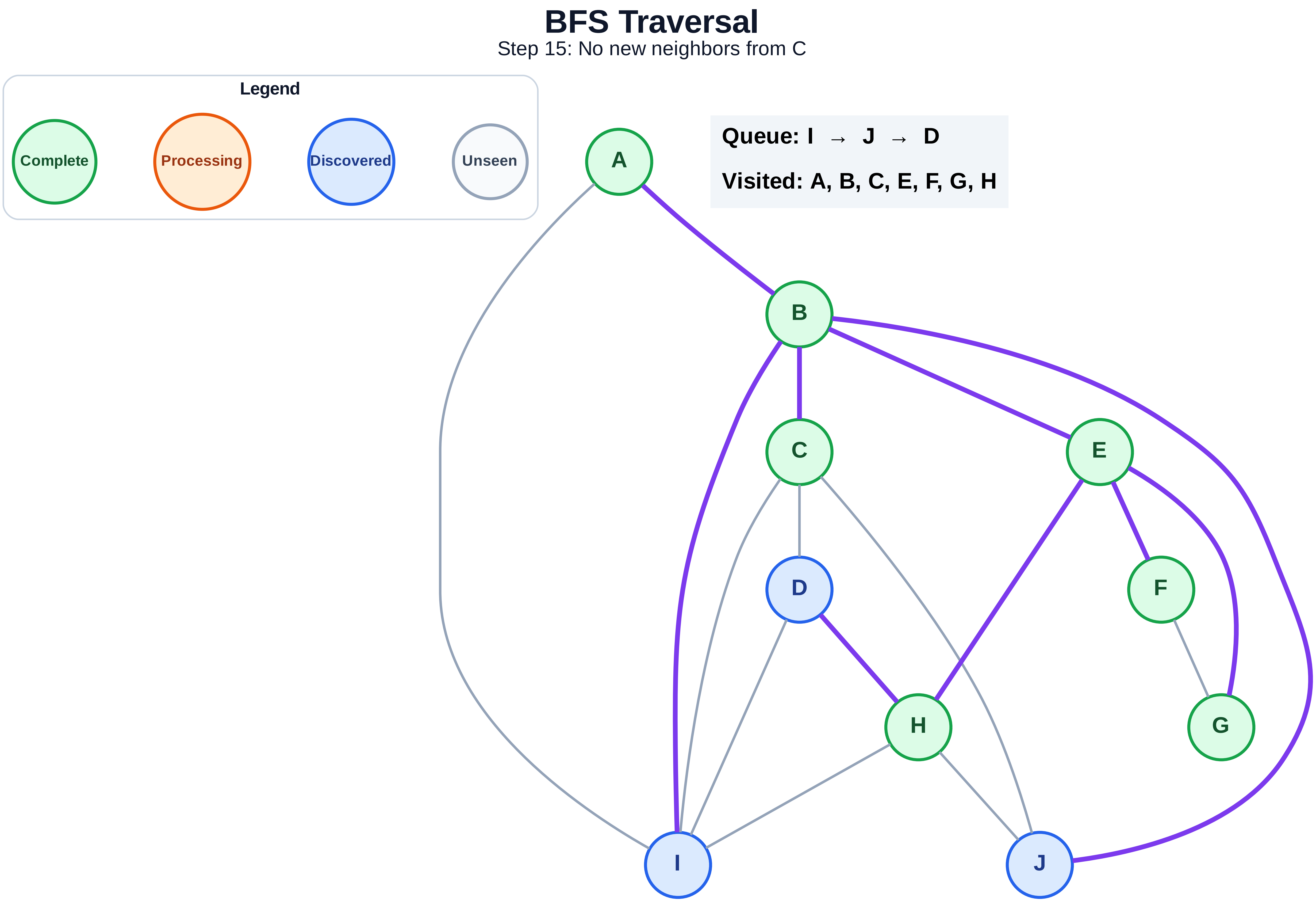
Step 13: No new neighbors from A



BFS Traversal

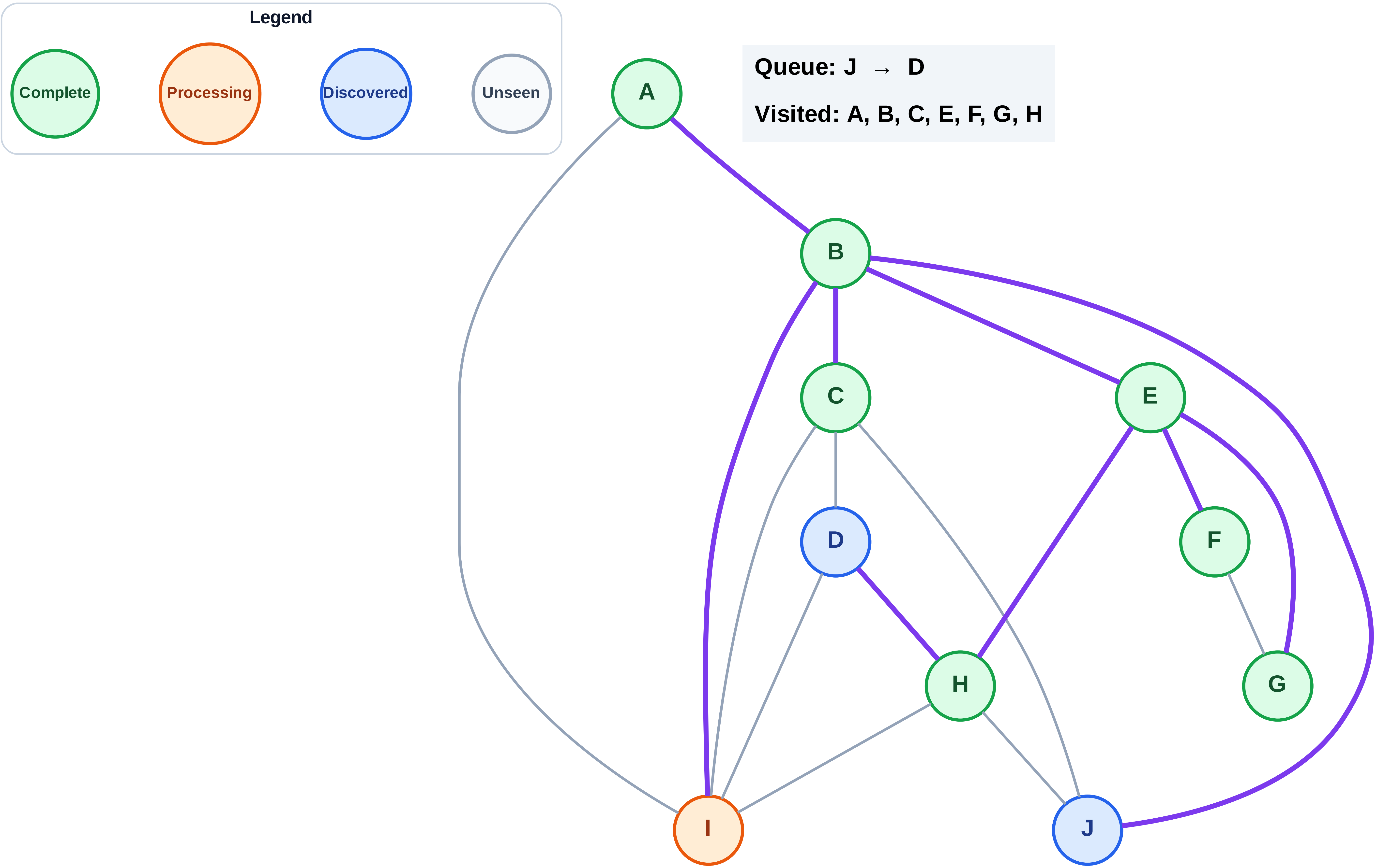
Step 14: Process node C




BFS Traversal
Step 15: No new neighbors from C
Legend
Complete
Processing
Discovered
Unseen
Queue: I → J → D
Visited: A, B, C, E, F, G, H
A
B
C
D
E
F
G
H
I
J

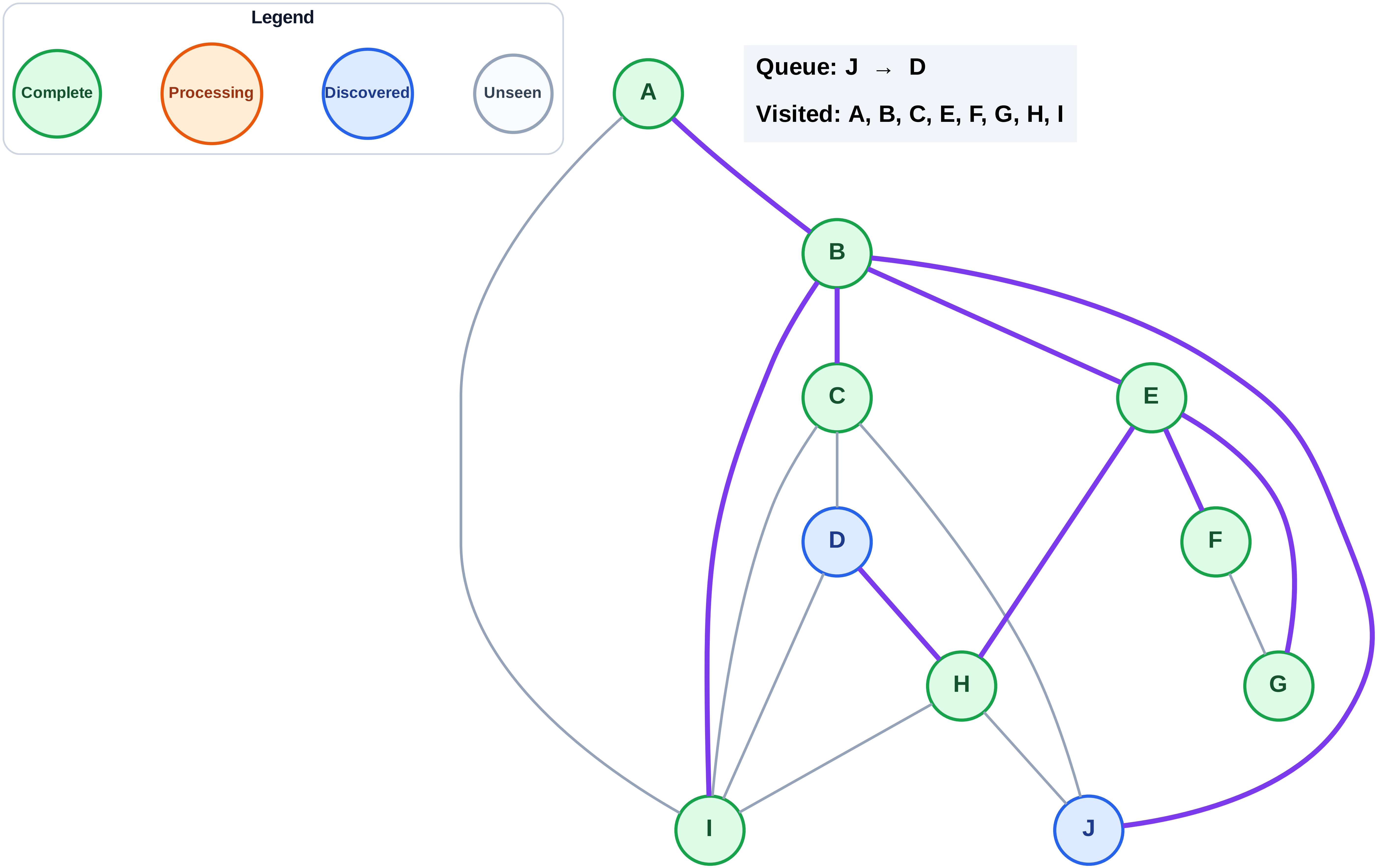
BFS Traversal

Step 16: Process node I



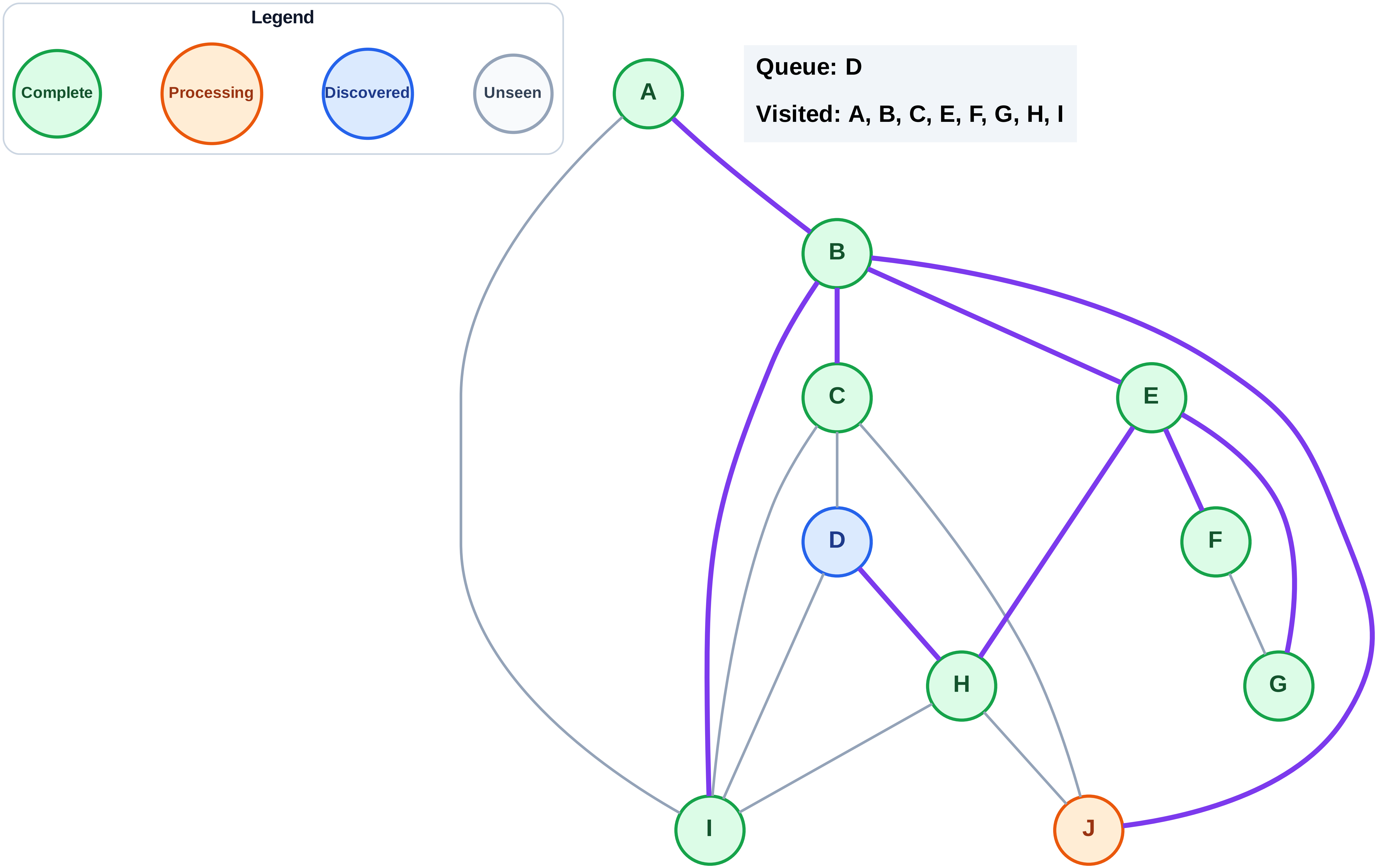
BFS Traversal

Step 17: No new neighbors from I



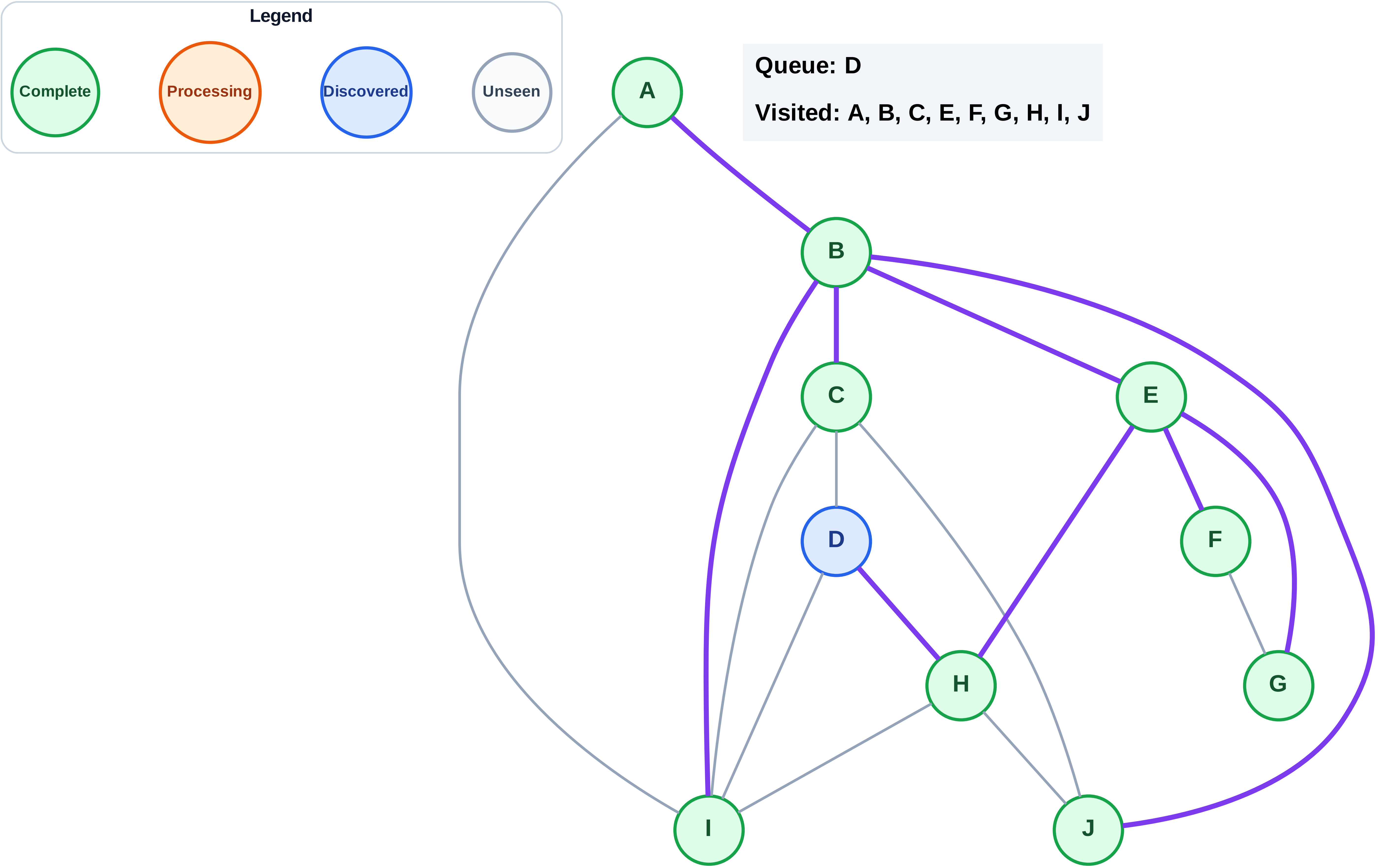
BFS Traversal

Step 18: Process node J



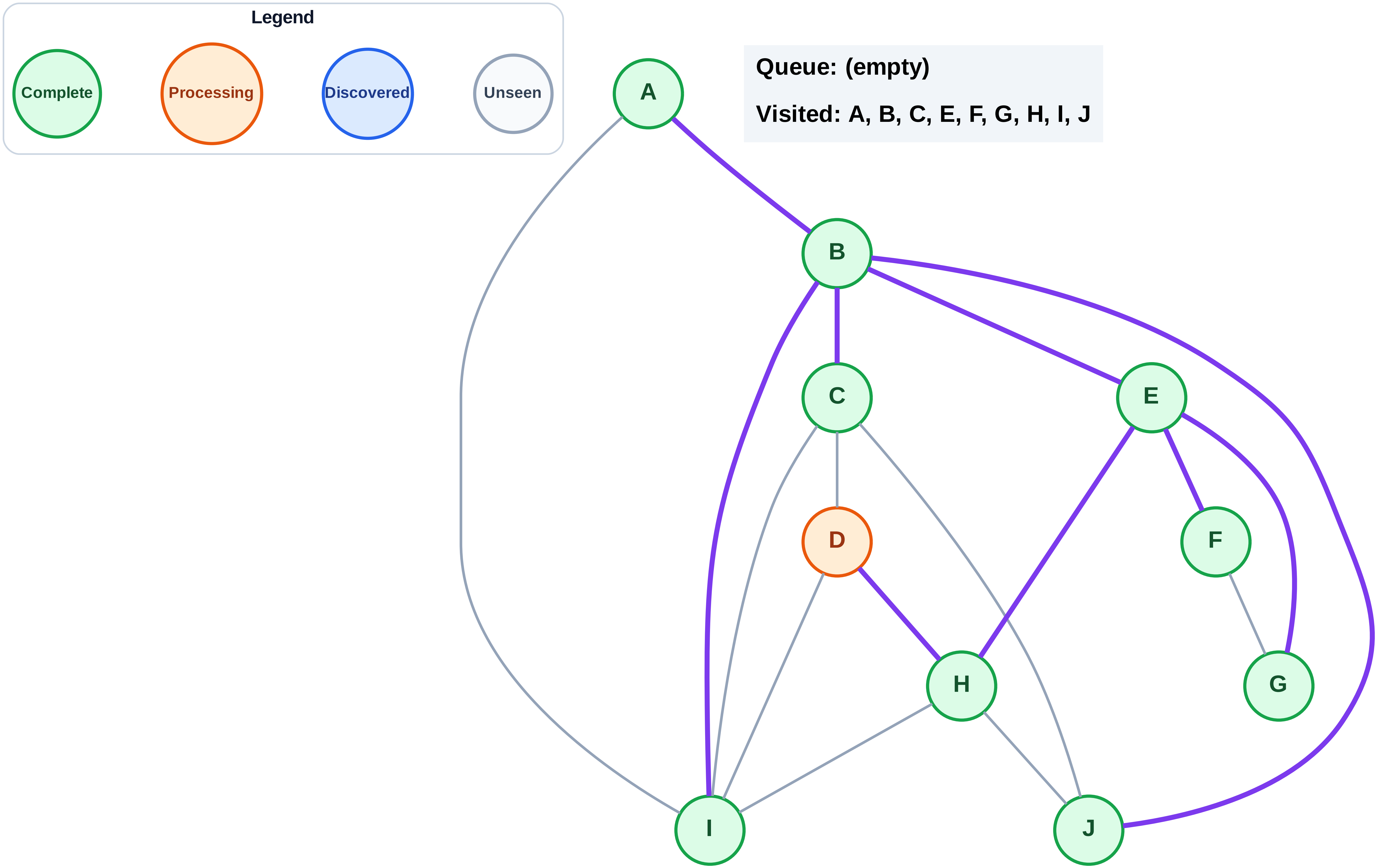
BFS Traversal

Step 19: No new neighbors from J



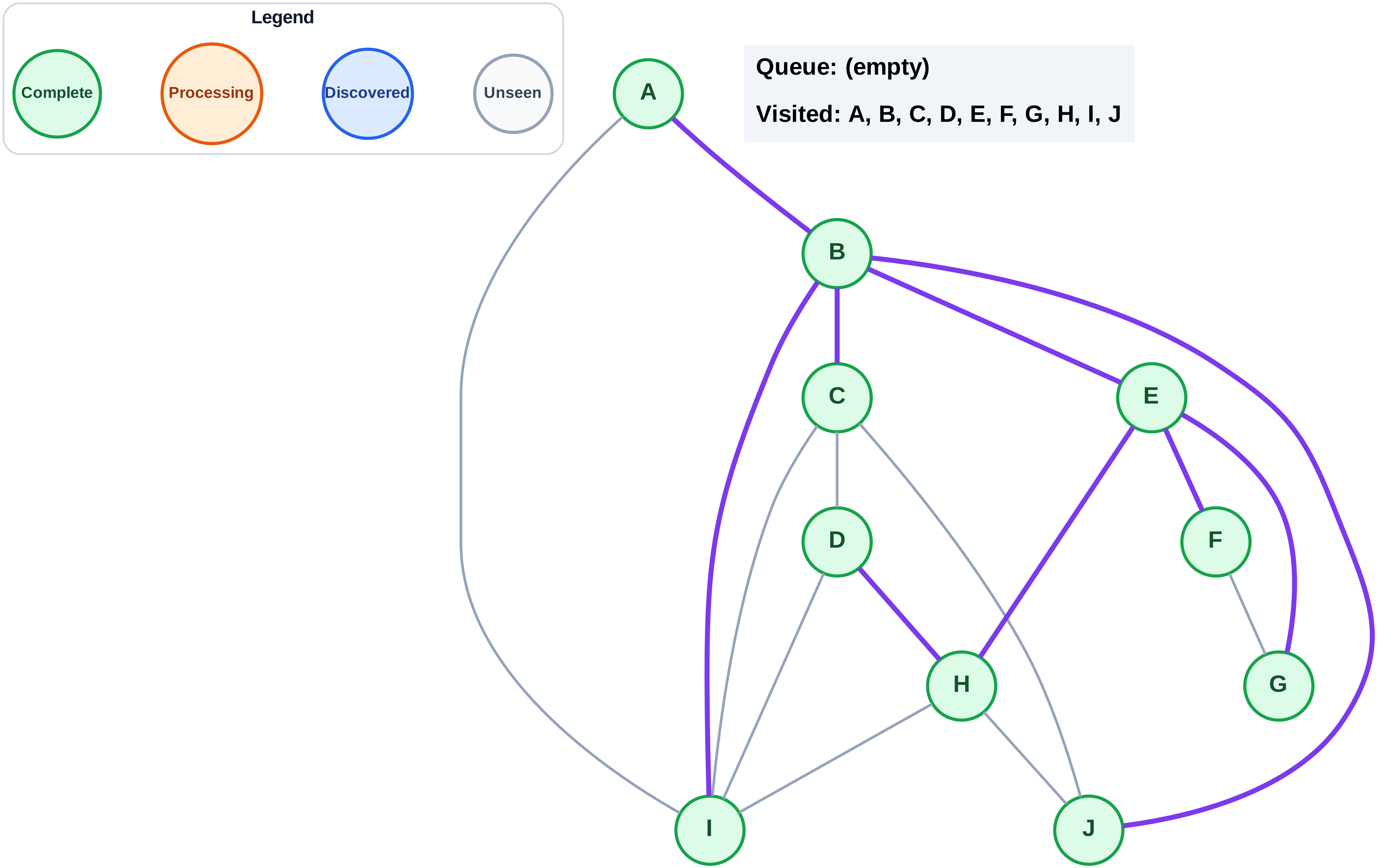
BFS Traversal

Step 20: Process node D



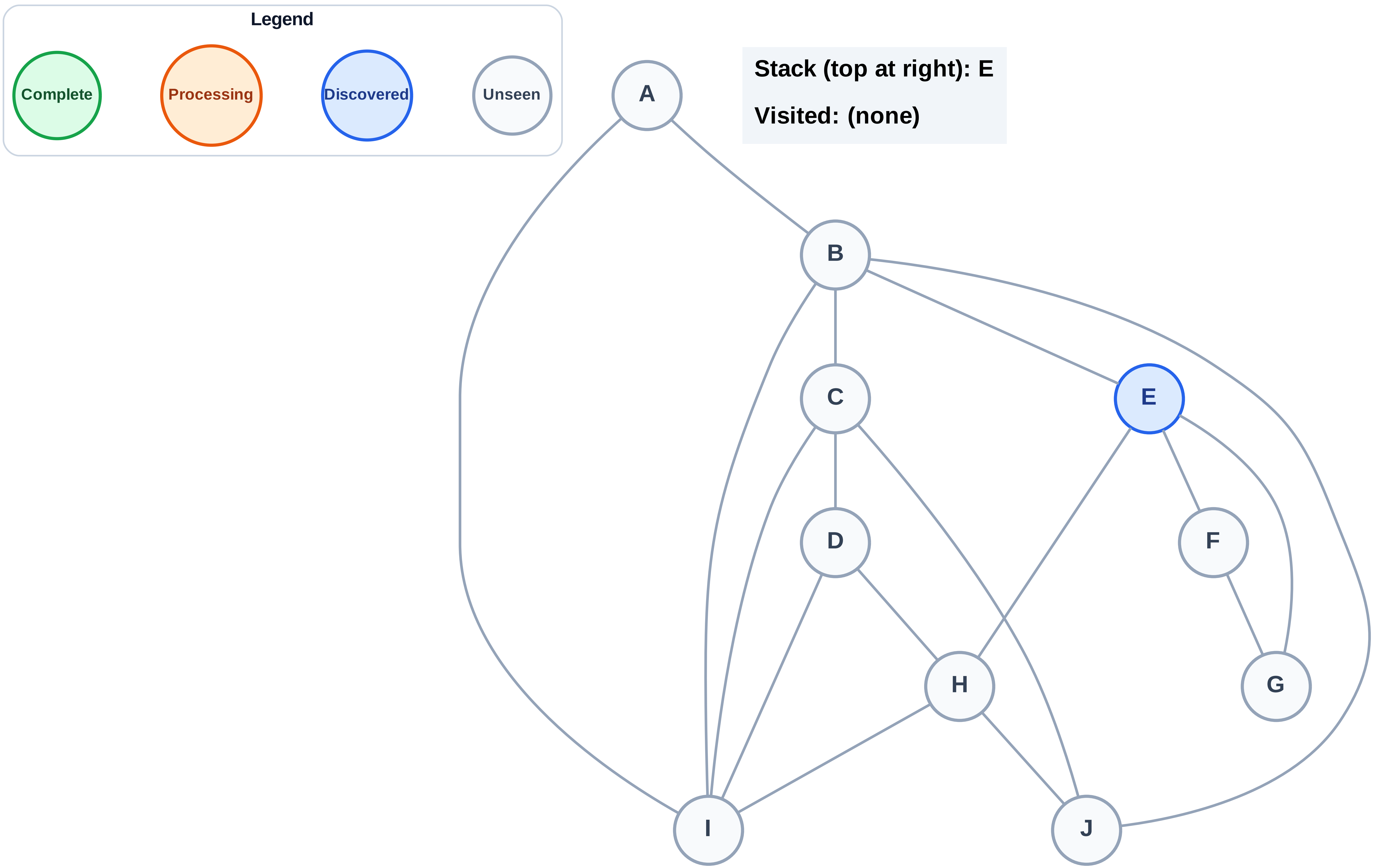
BFS Traversal

Step 21: No new neighbors from D



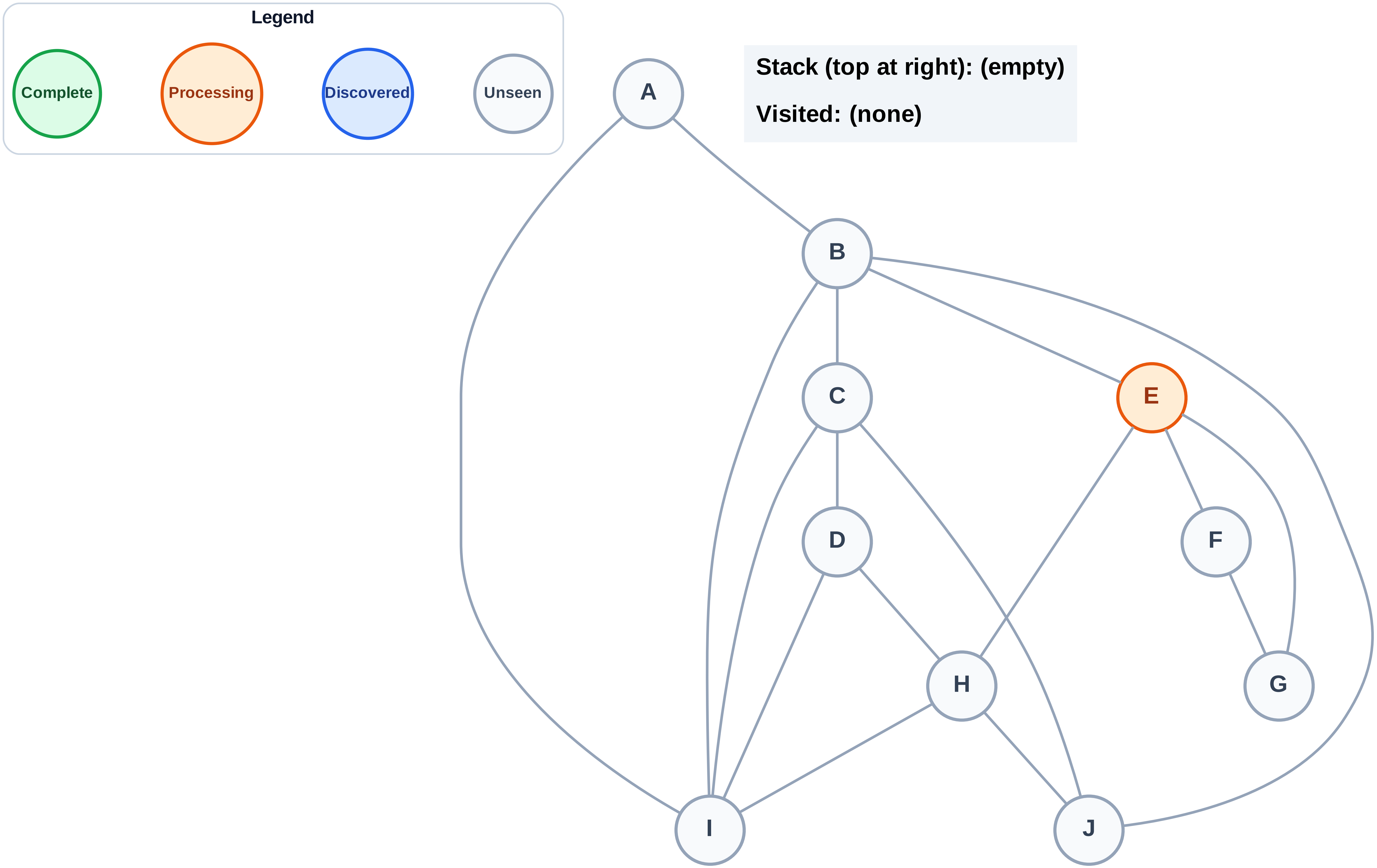
DFS Traversal

Step 1: Start at E



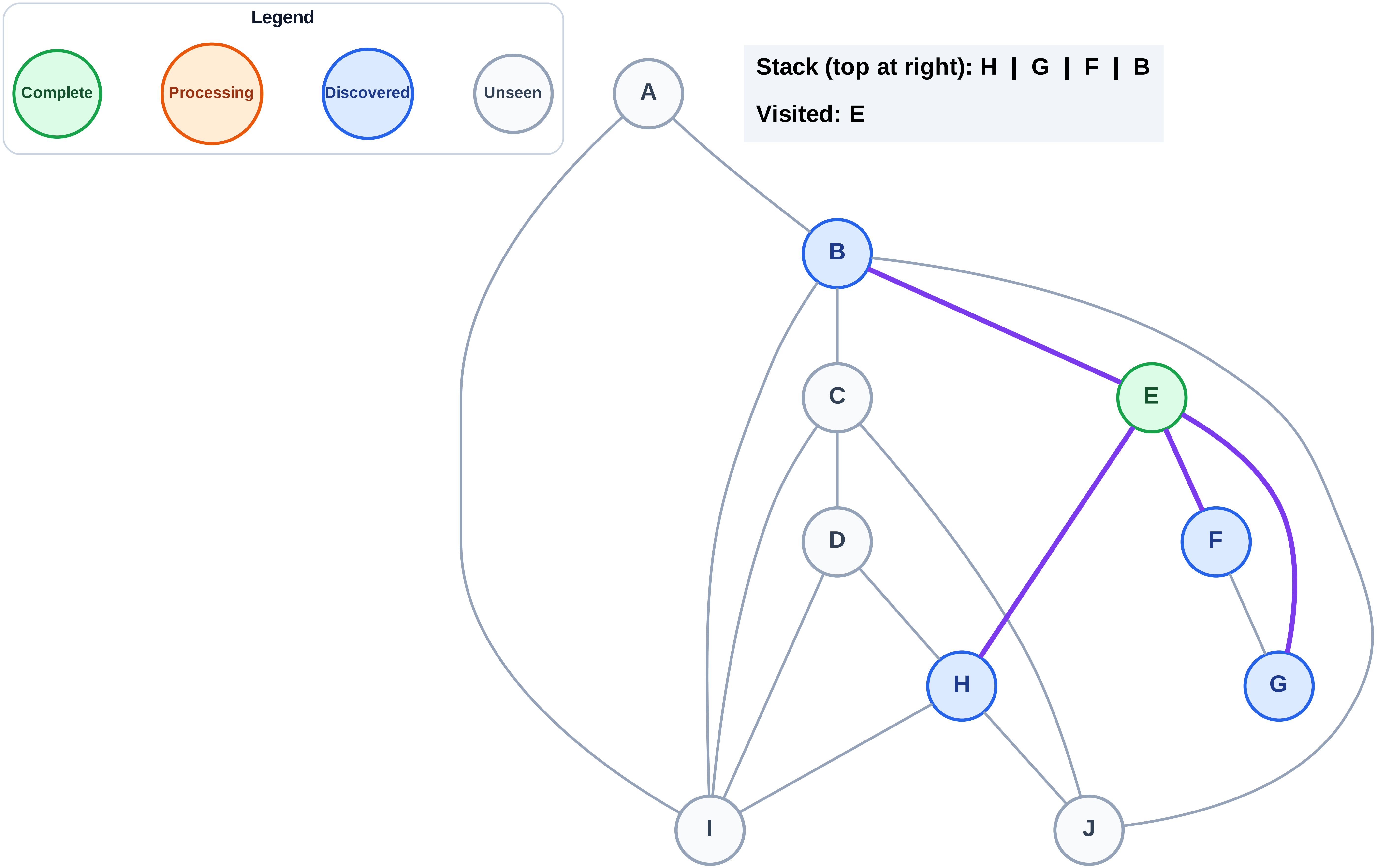
DFS Traversal

Step 2: Process node E



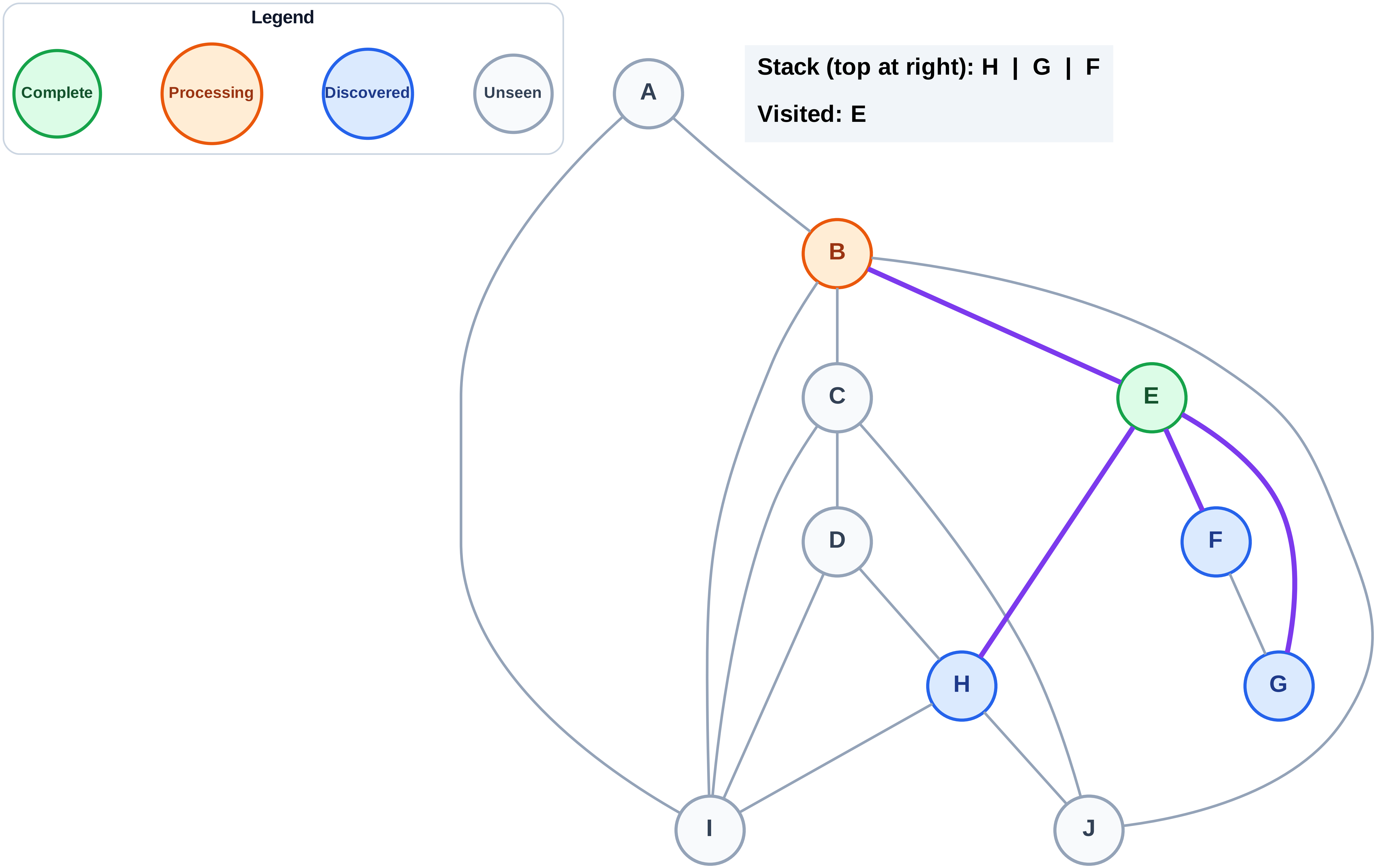
DFS Traversal

Step 3: Discover H, G, F, B from E



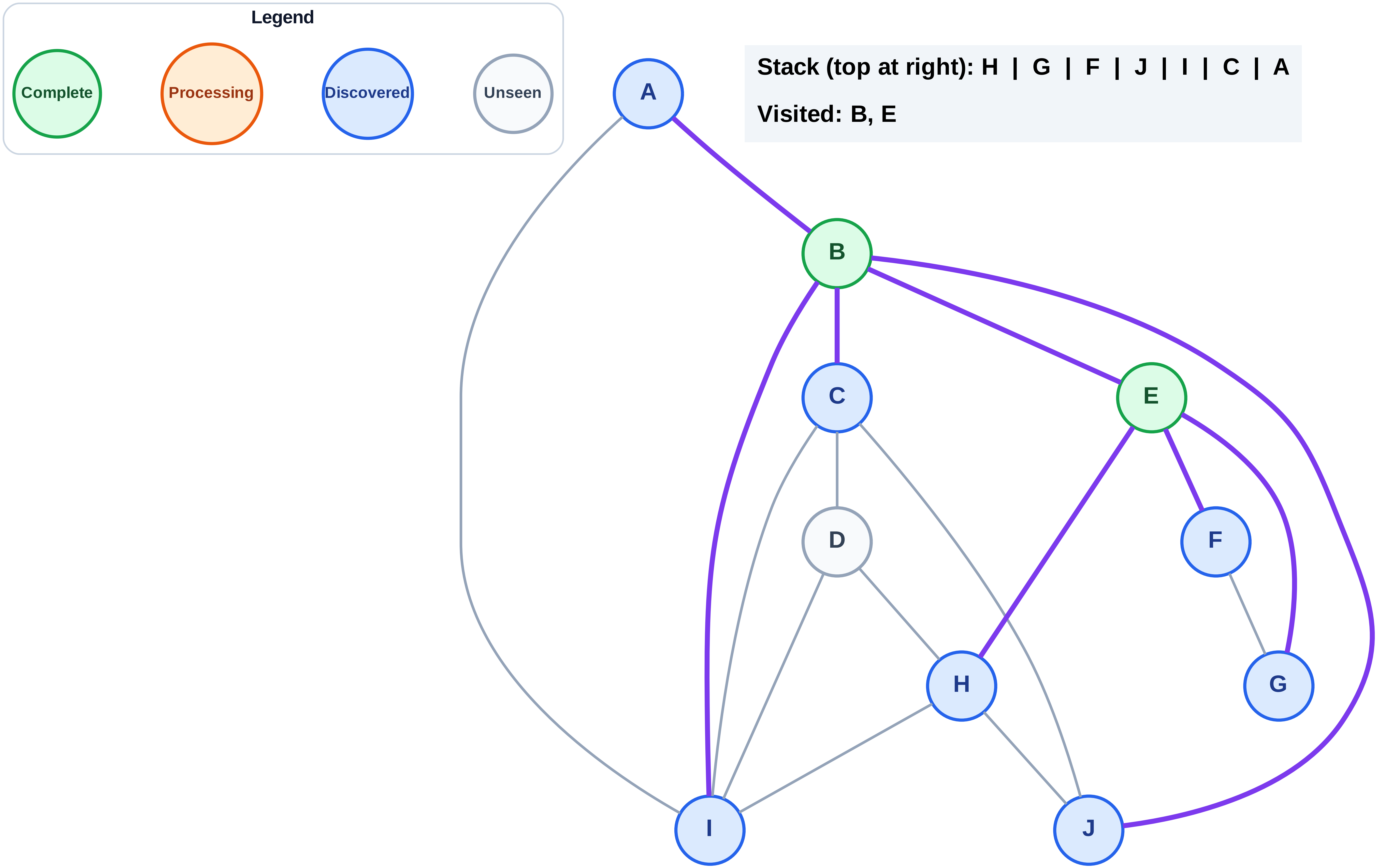
DFS Traversal

Step 4: Process node B



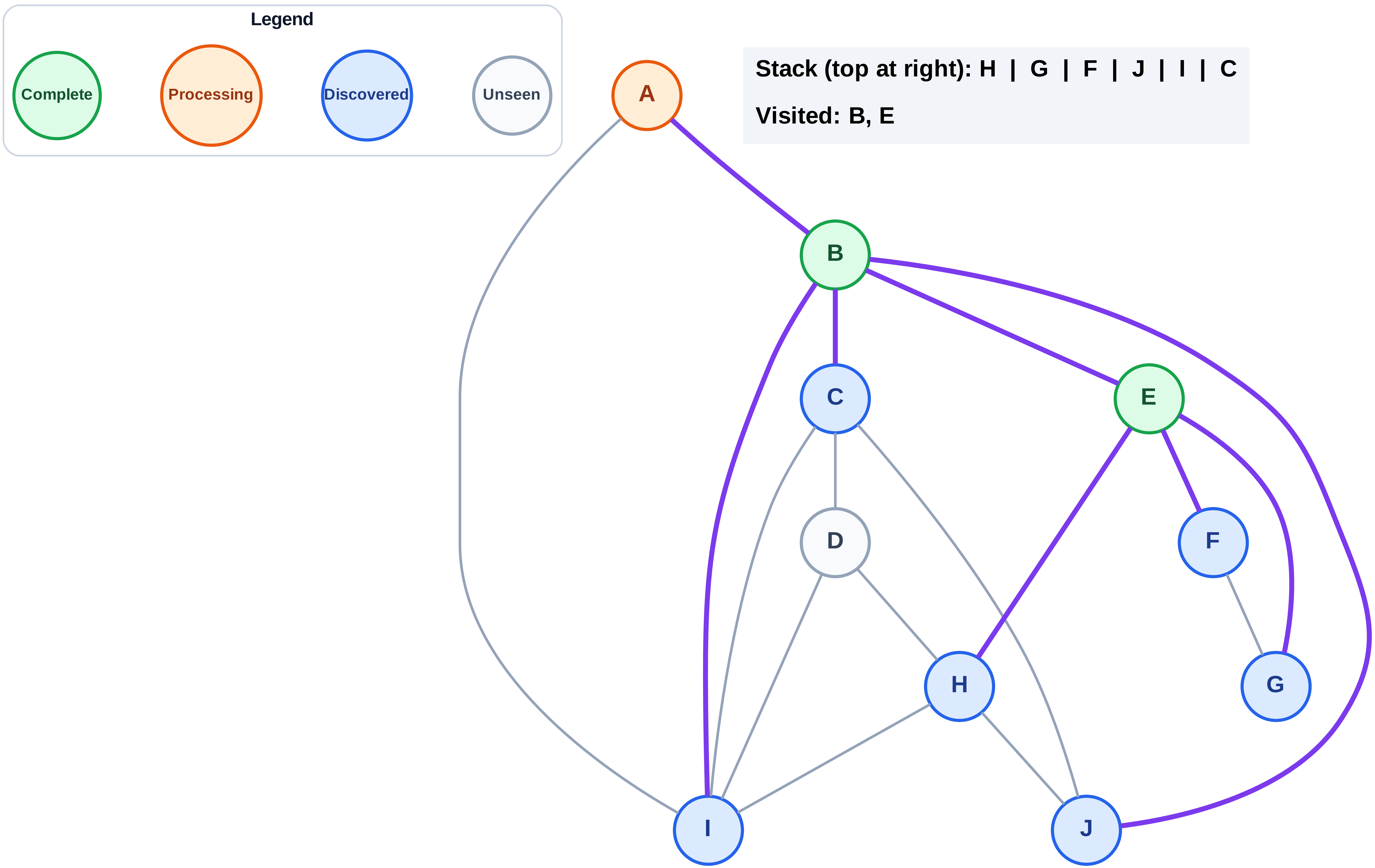
DFS Traversal

Step 5: Discover J, I, C, A from B



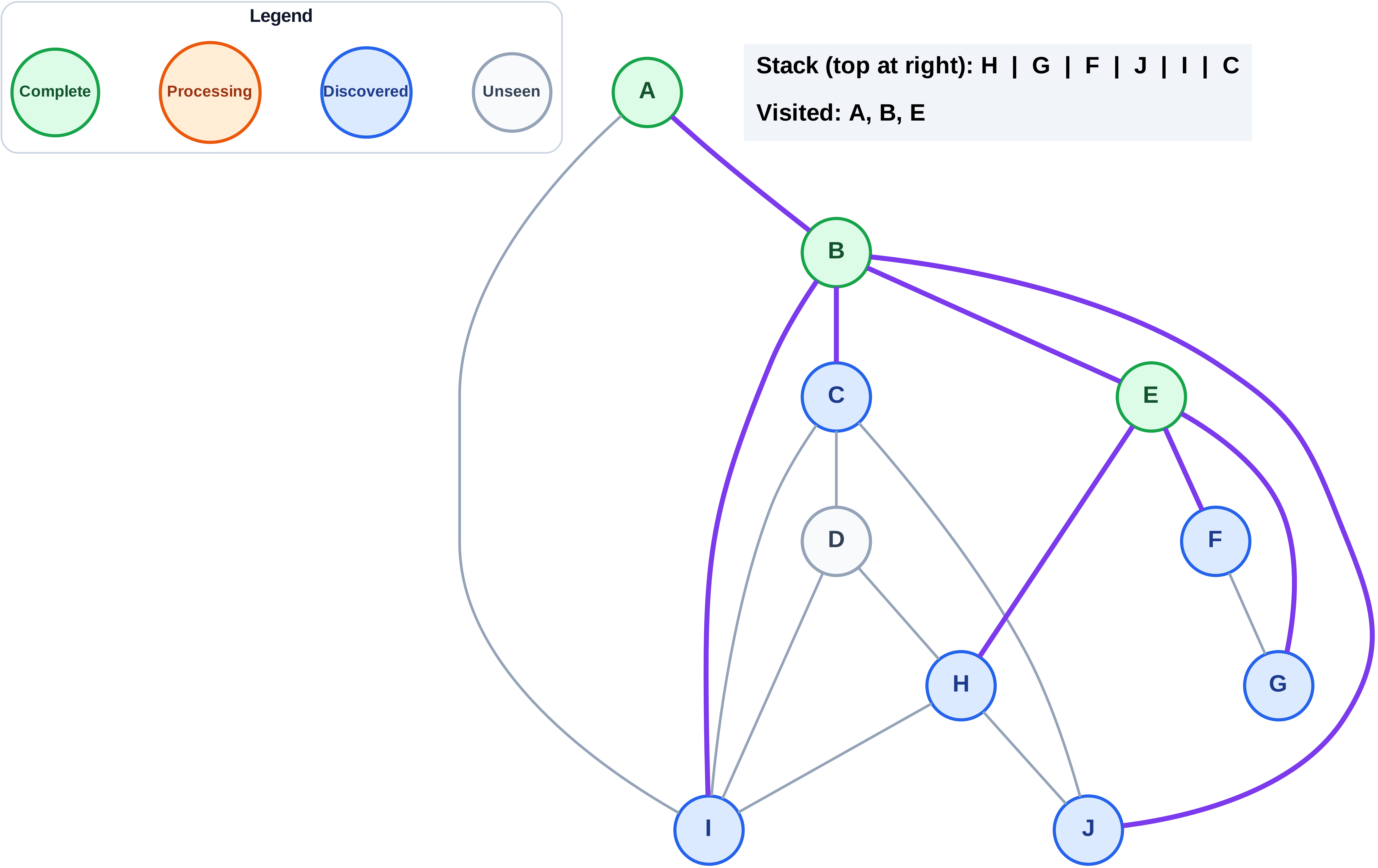
DFS Traversal

Step 6: Process node A



DFS Traversal

Step 7: No new neighbors from A

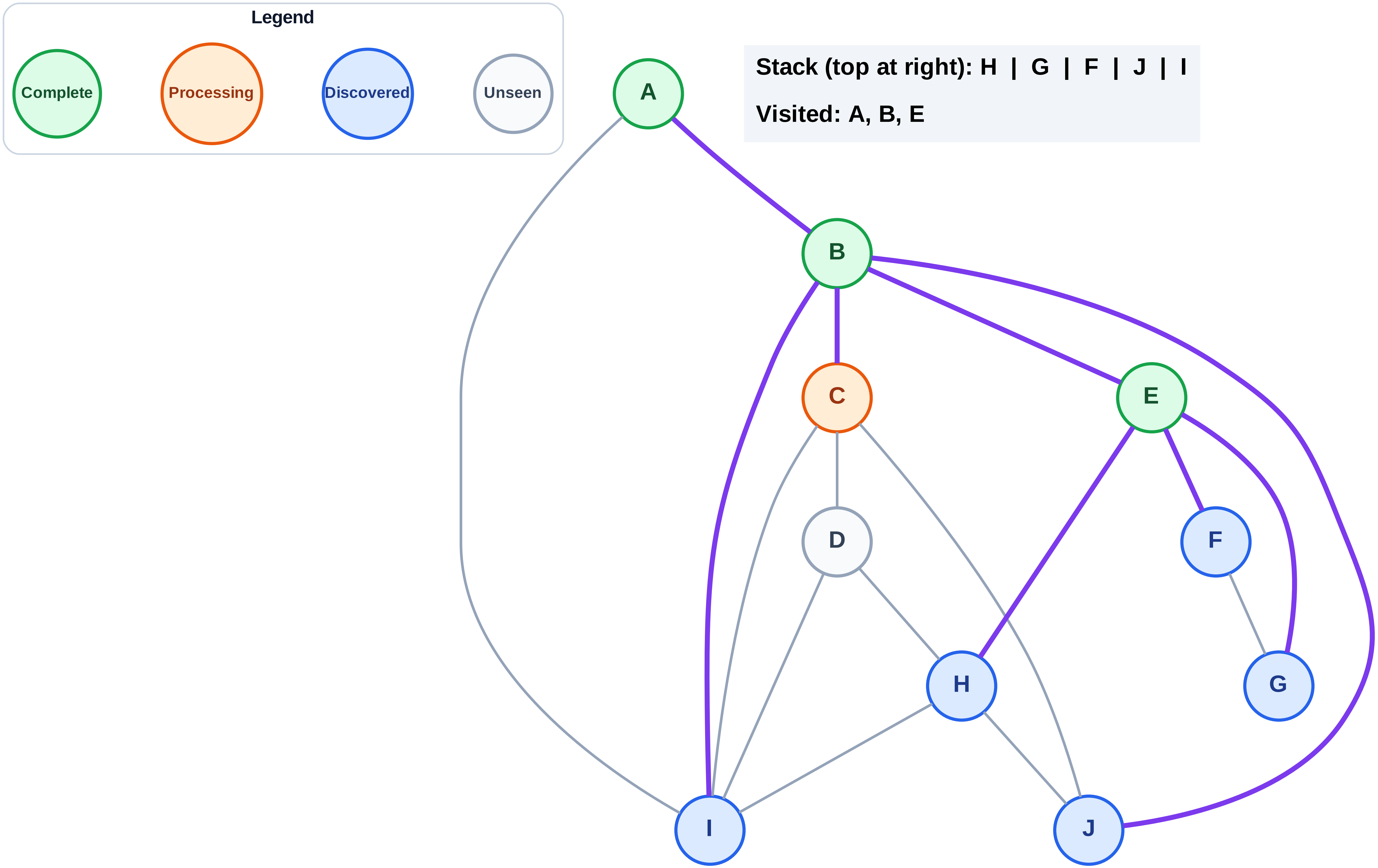


Stack (top at right): H | G | F | J | I | C

Visited: A, B, E

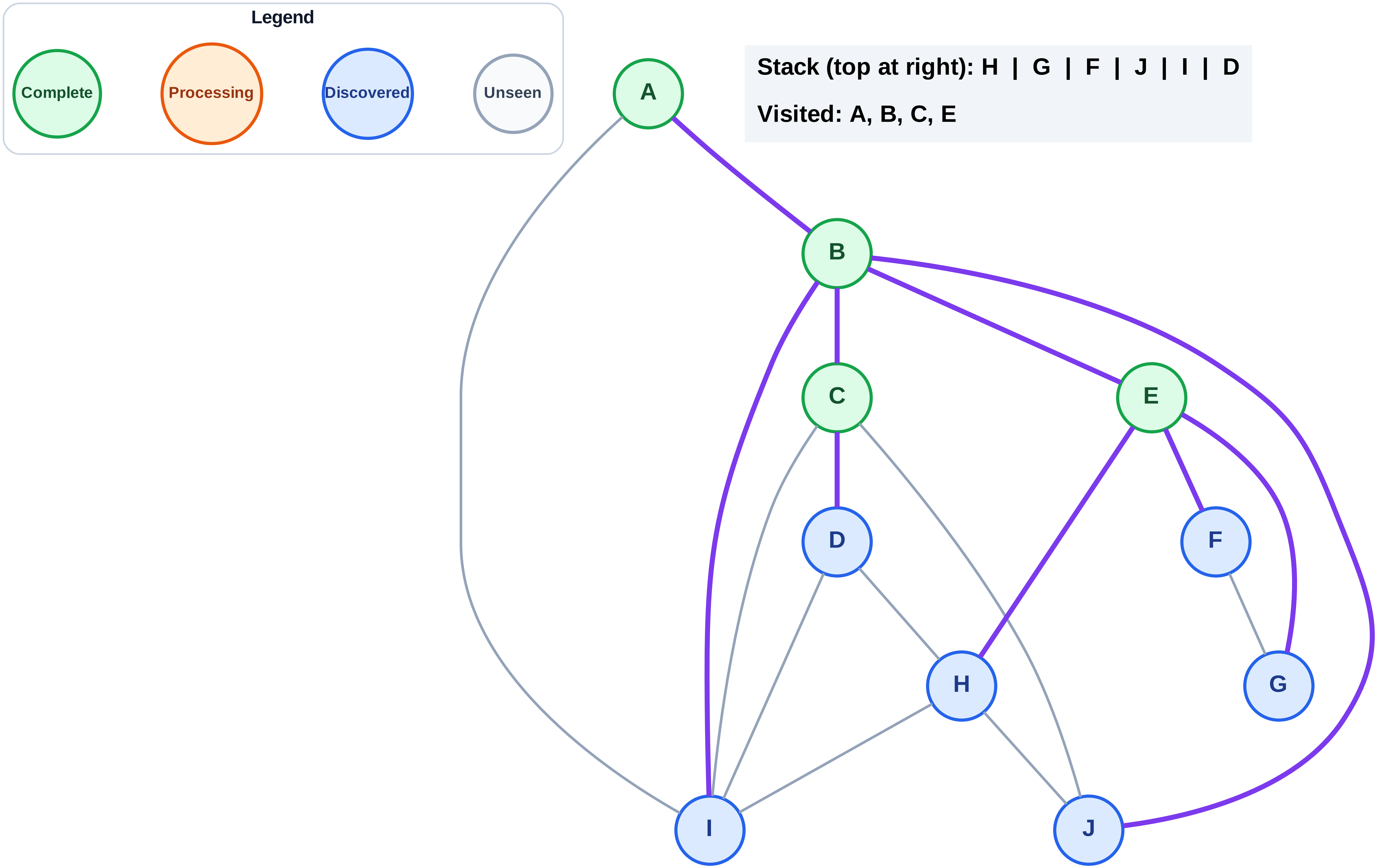
DFS Traversal

Step 8: Process node C



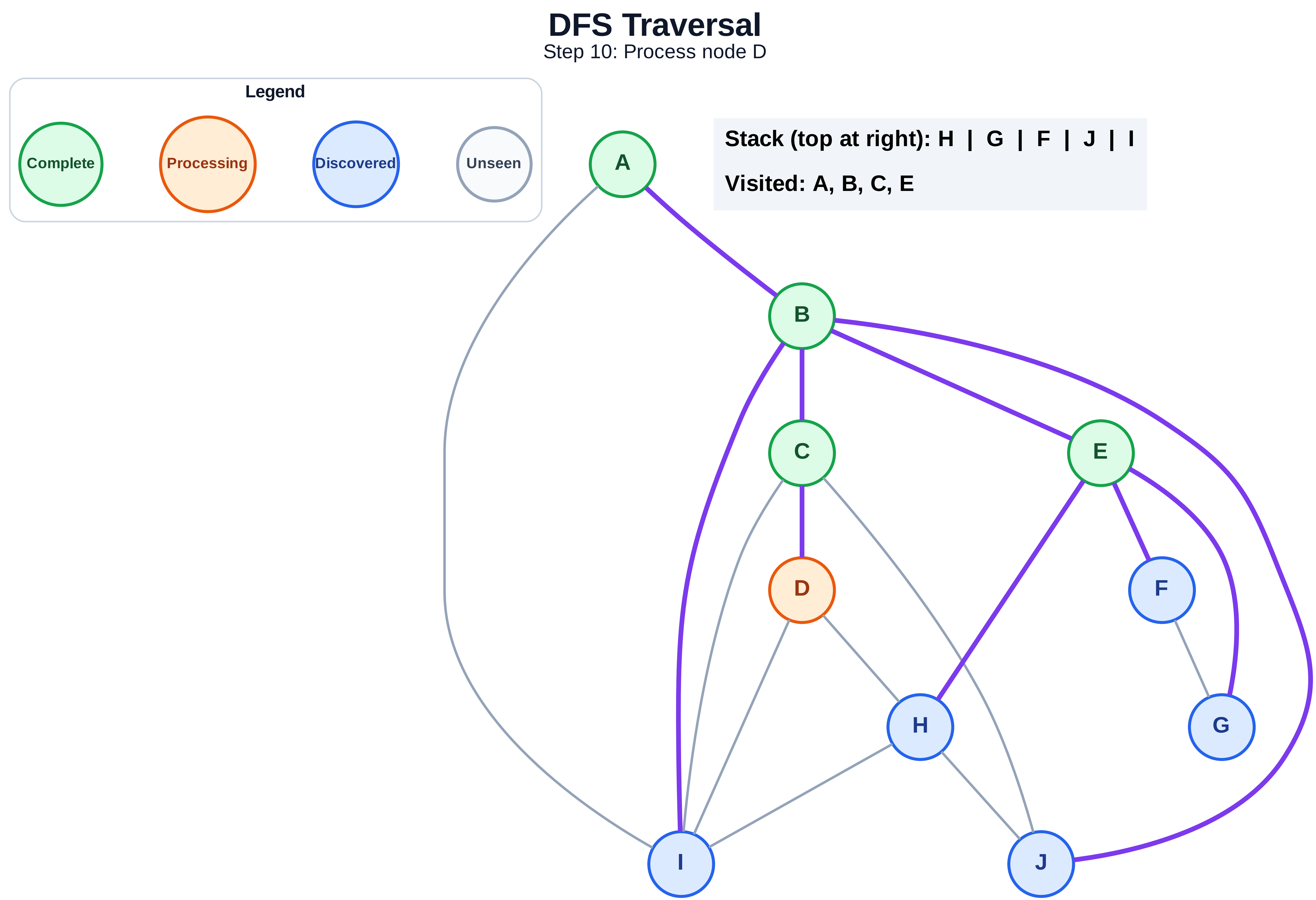
DFS Traversal

Step 9: Discover D from C



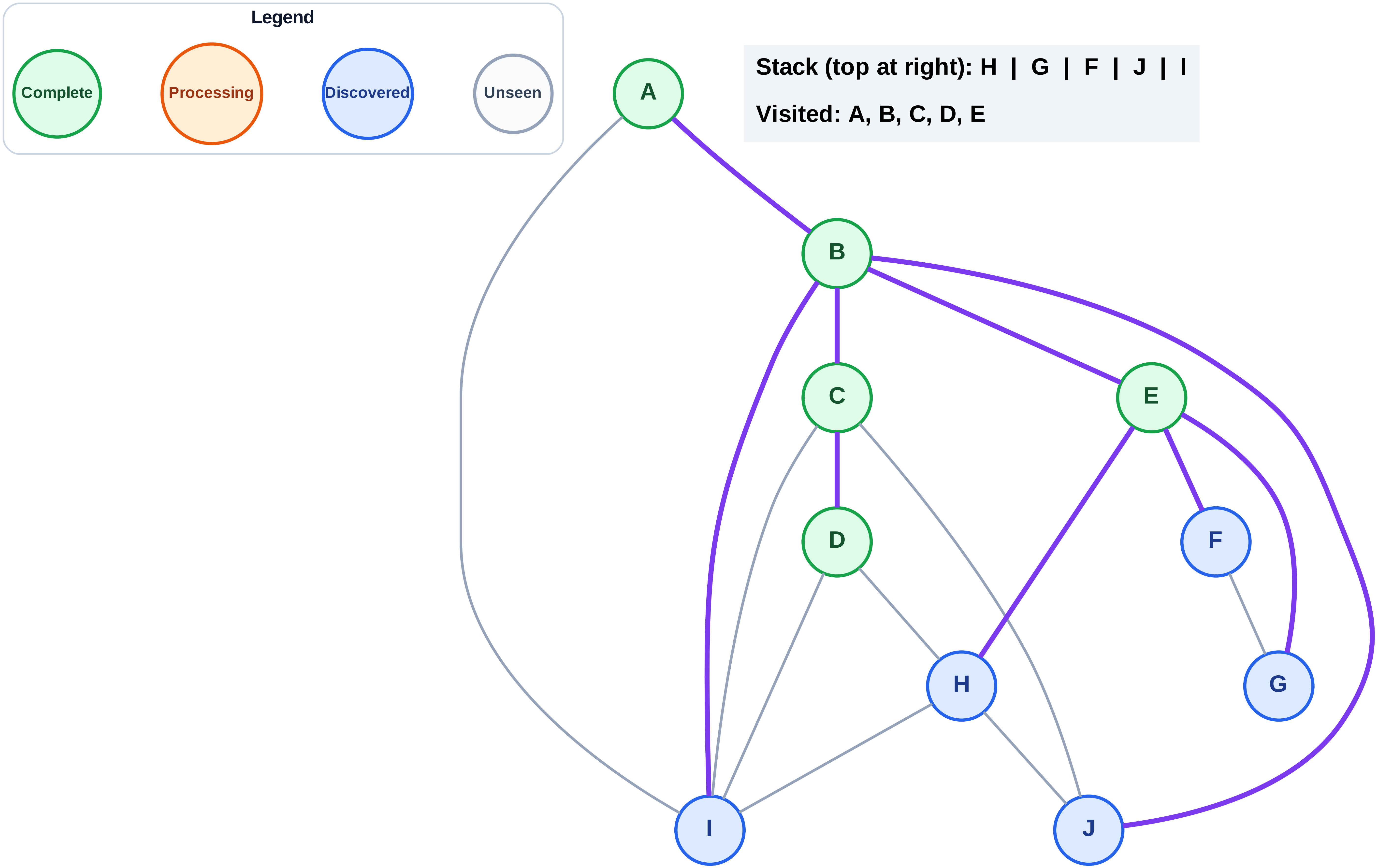
Stack (top at right): H | G | F | J | I | D

Visited: A, B, C, E


DFS Traversal
Step 10: Process node D
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | G | F | J | I
Visited: A, B, C, E
A
B
C
D
E
F
G
H
I
J

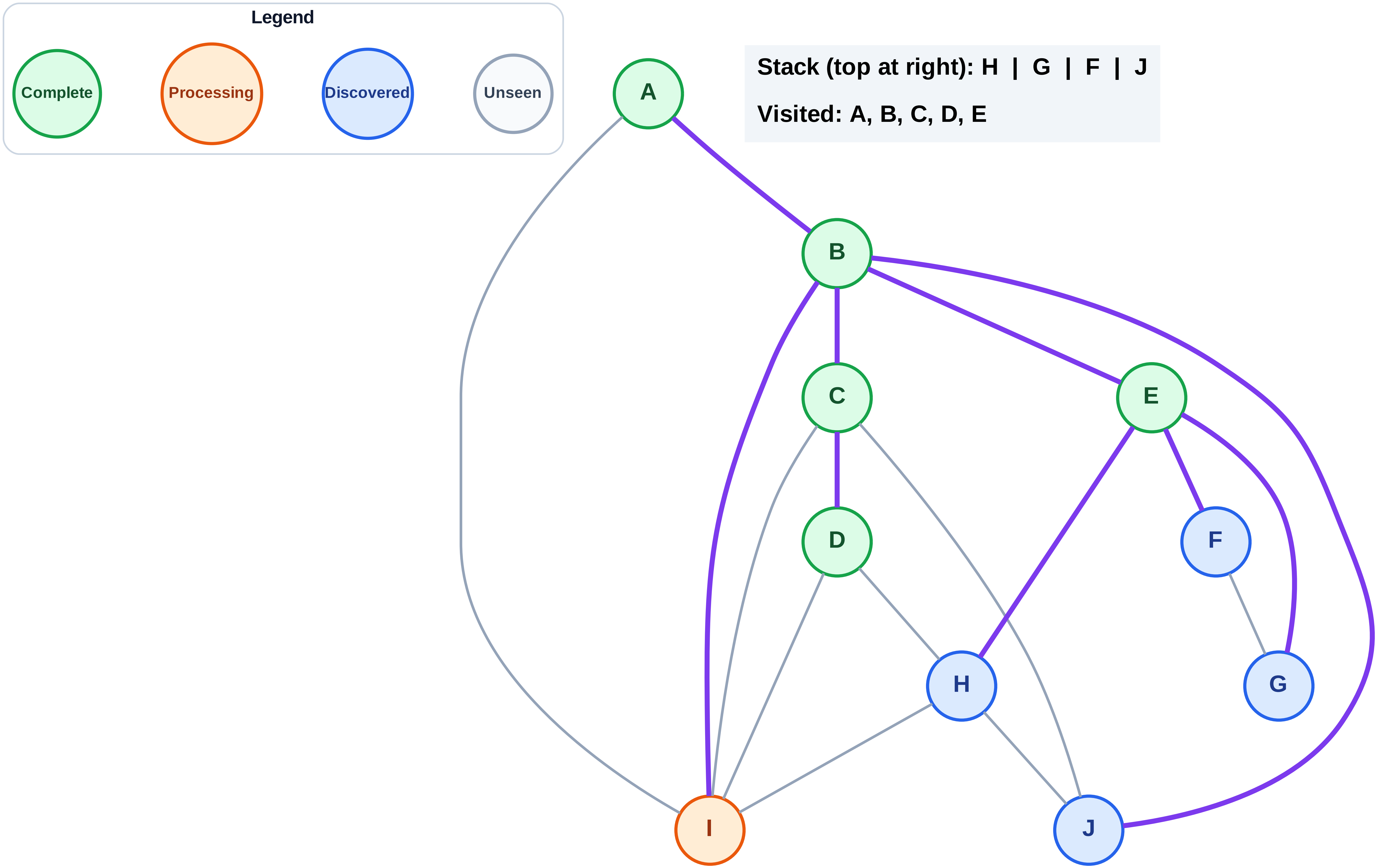
DFS Traversal

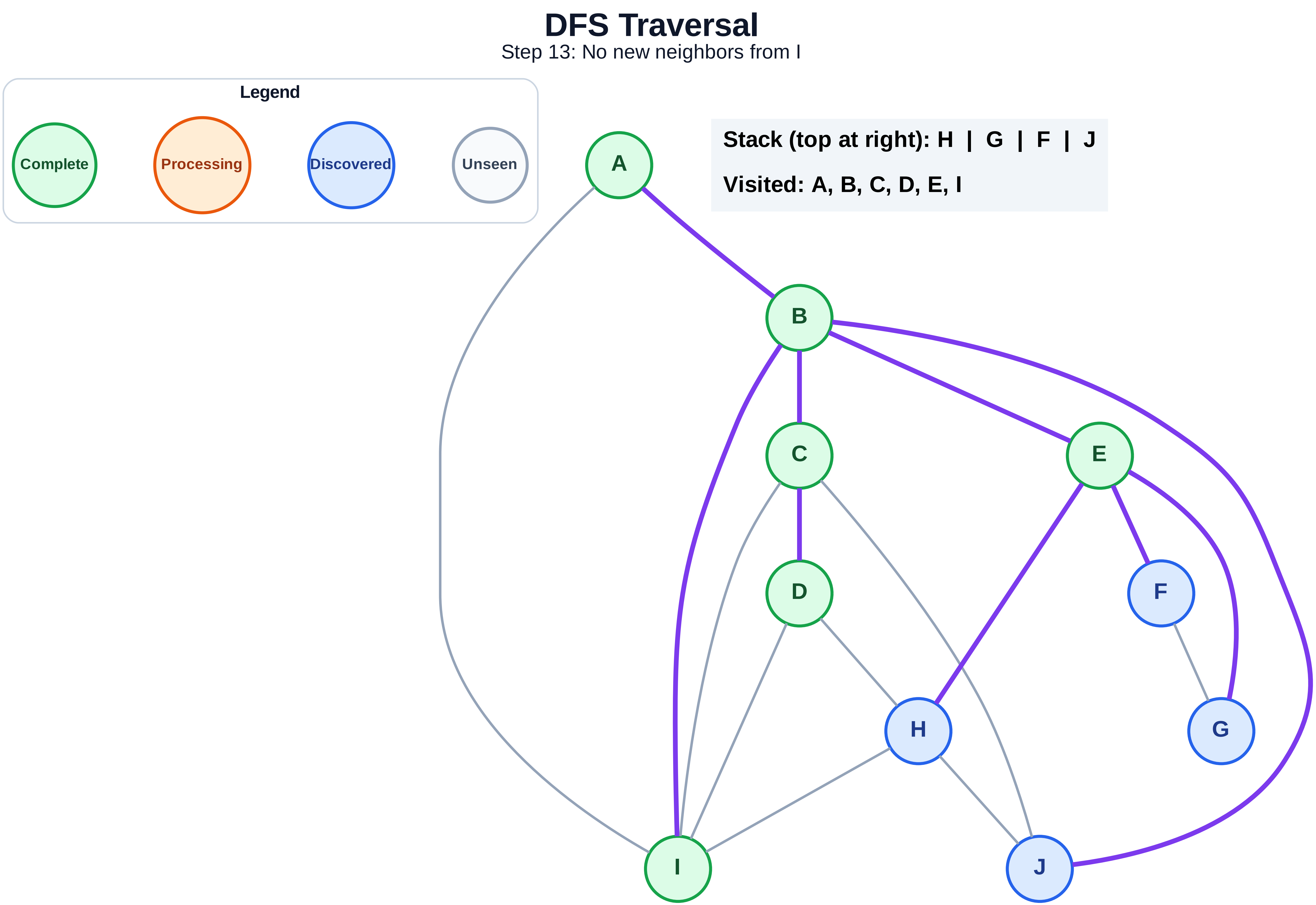
Step 11: No new neighbors from D



DFS Traversal

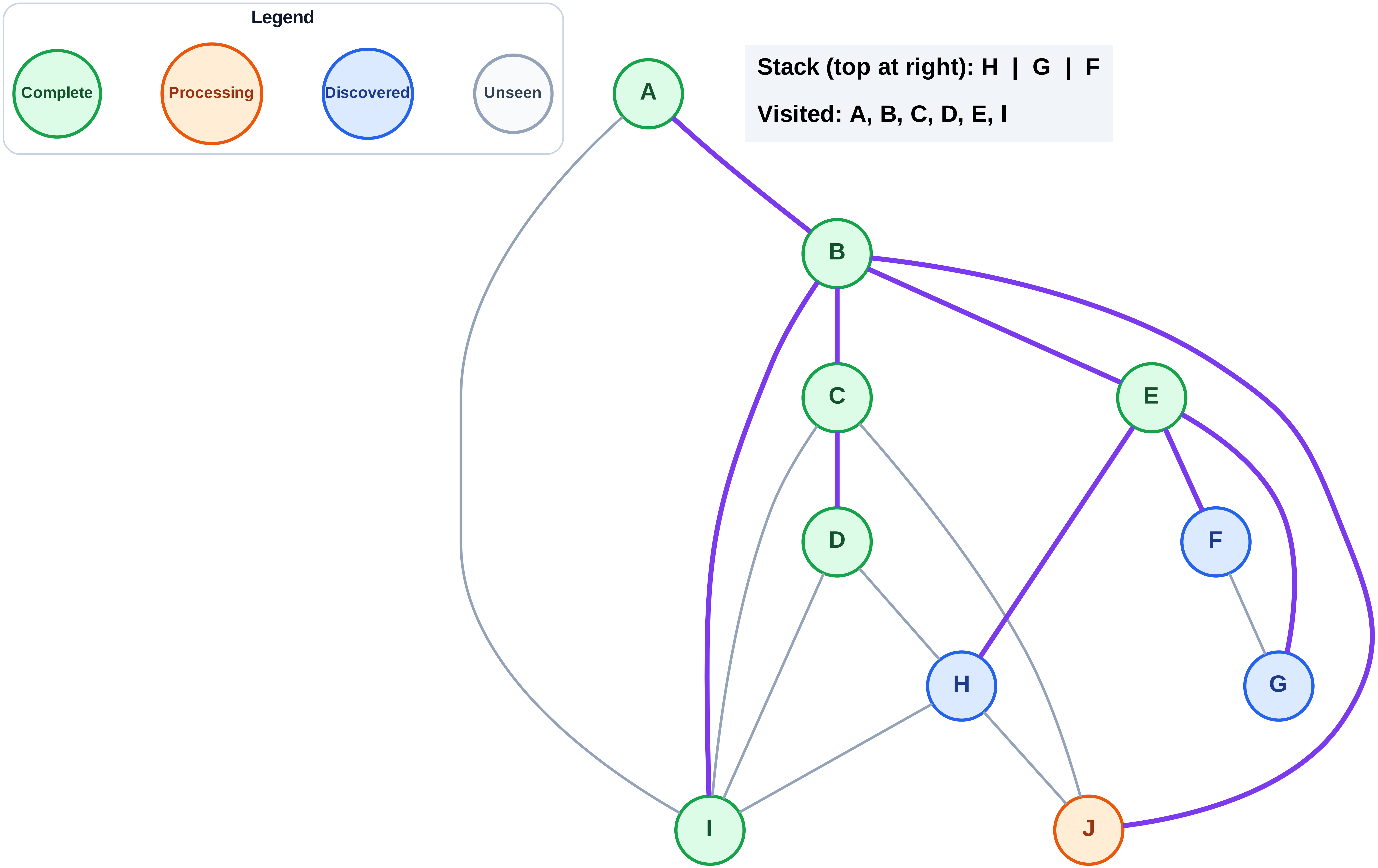
Step 12: Process node I




DFS Traversal
Step 13: No new neighbors from I
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | G | F | J
Visited: A, B, C, D, E, I
A
B
C
D
E
F
G
H
I
J

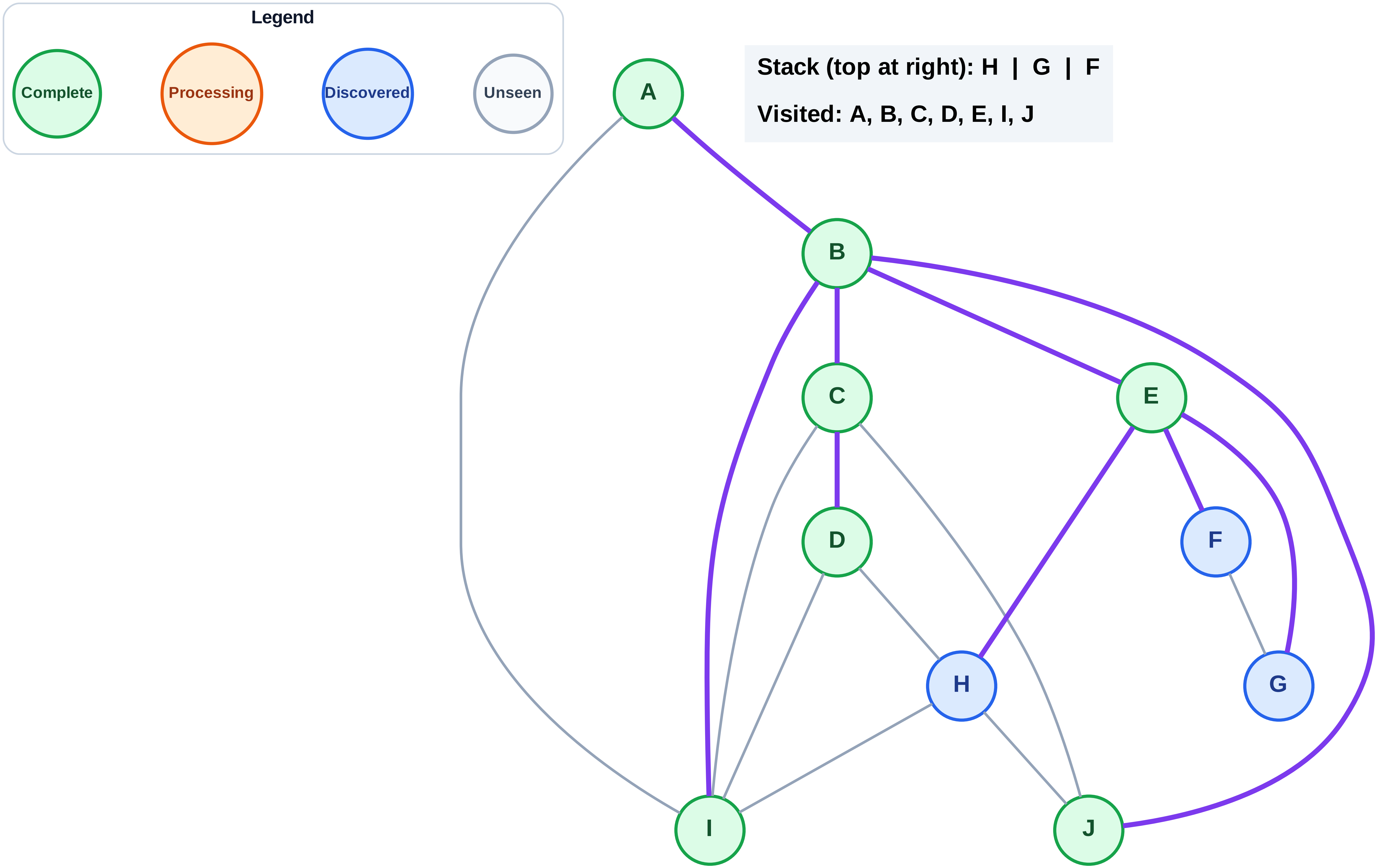
DFS Traversal

Step 14: Process node J



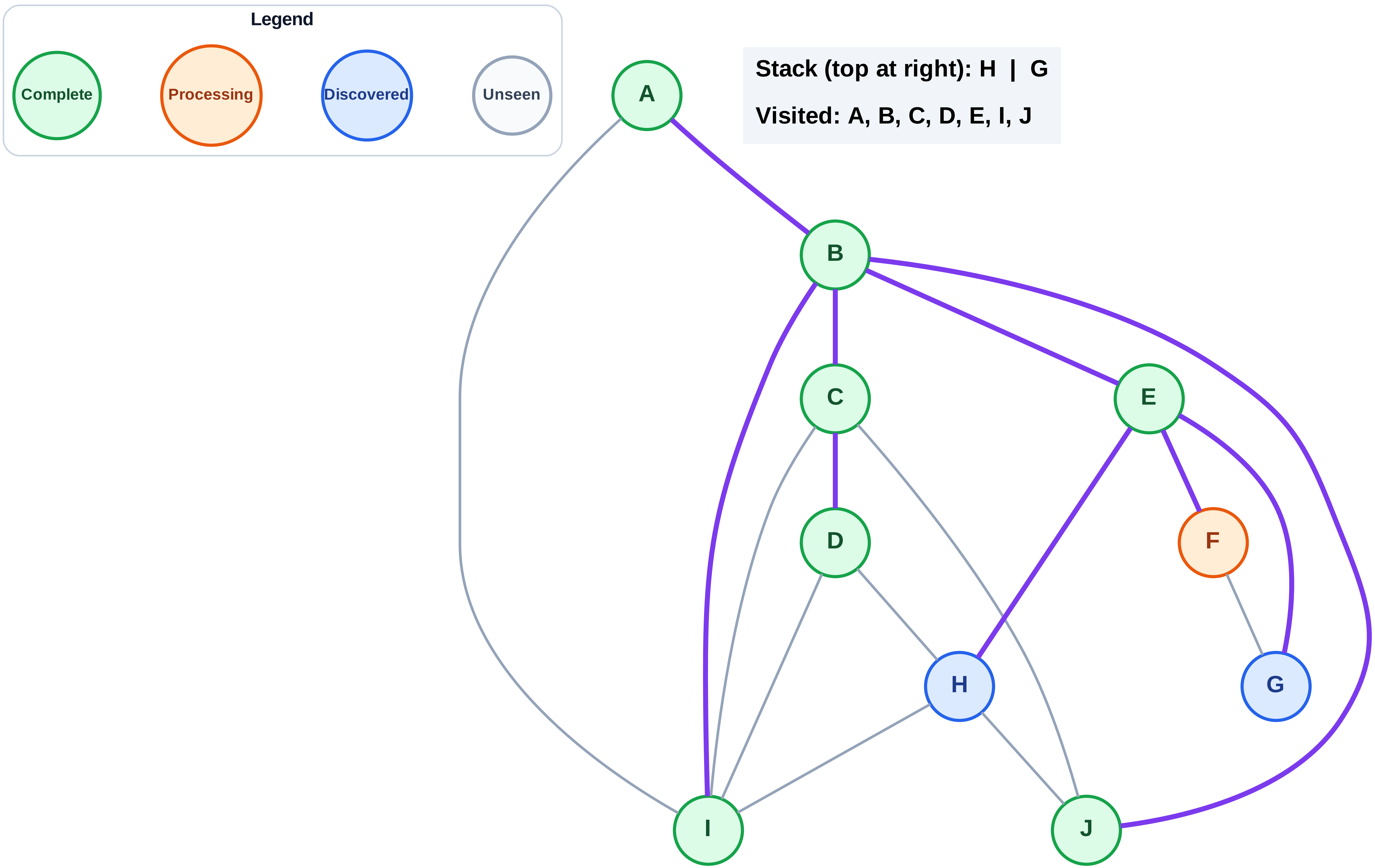
DFS Traversal

Step 15: No new neighbors from J



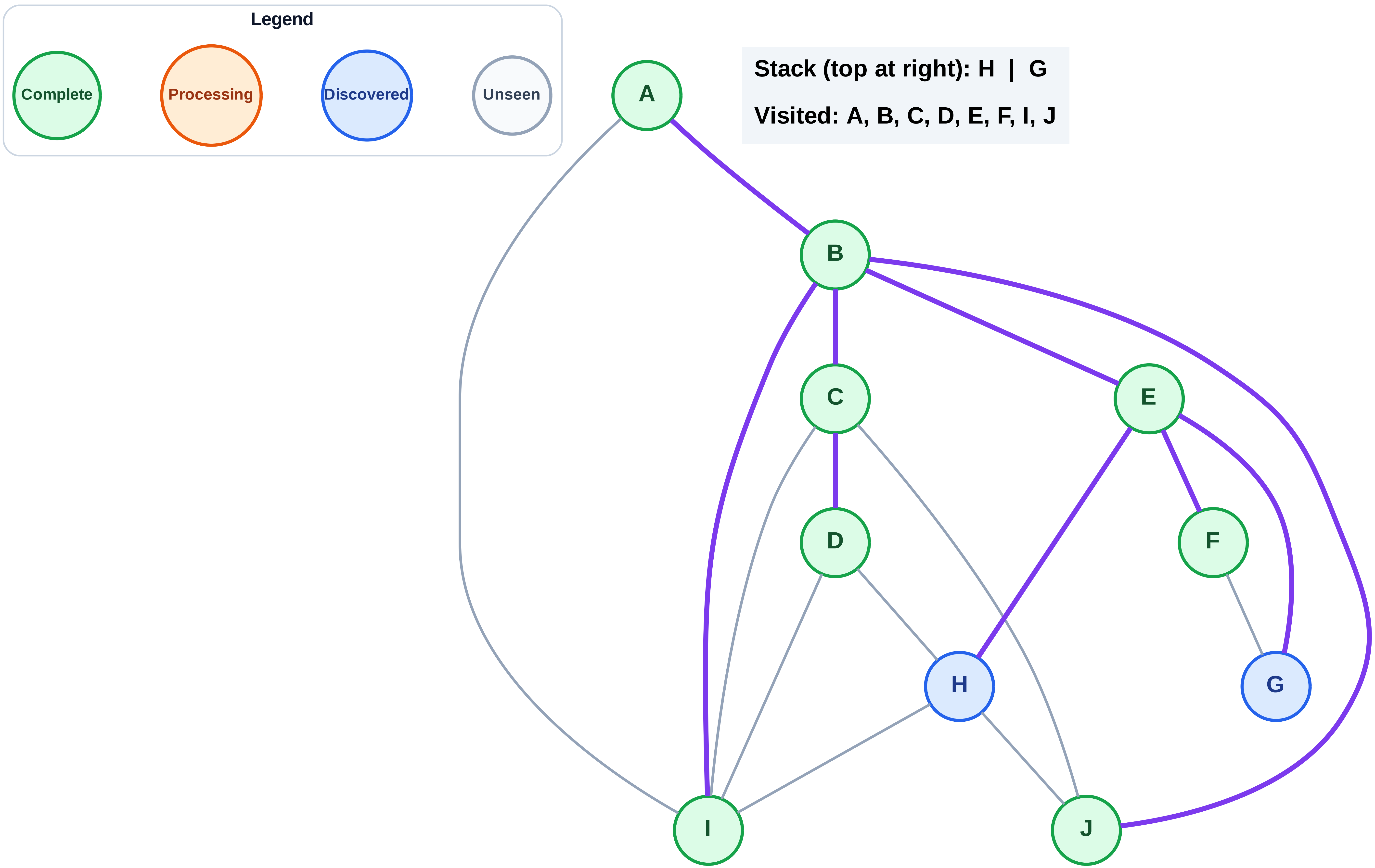
DFS Traversal

Step 16: Process node F



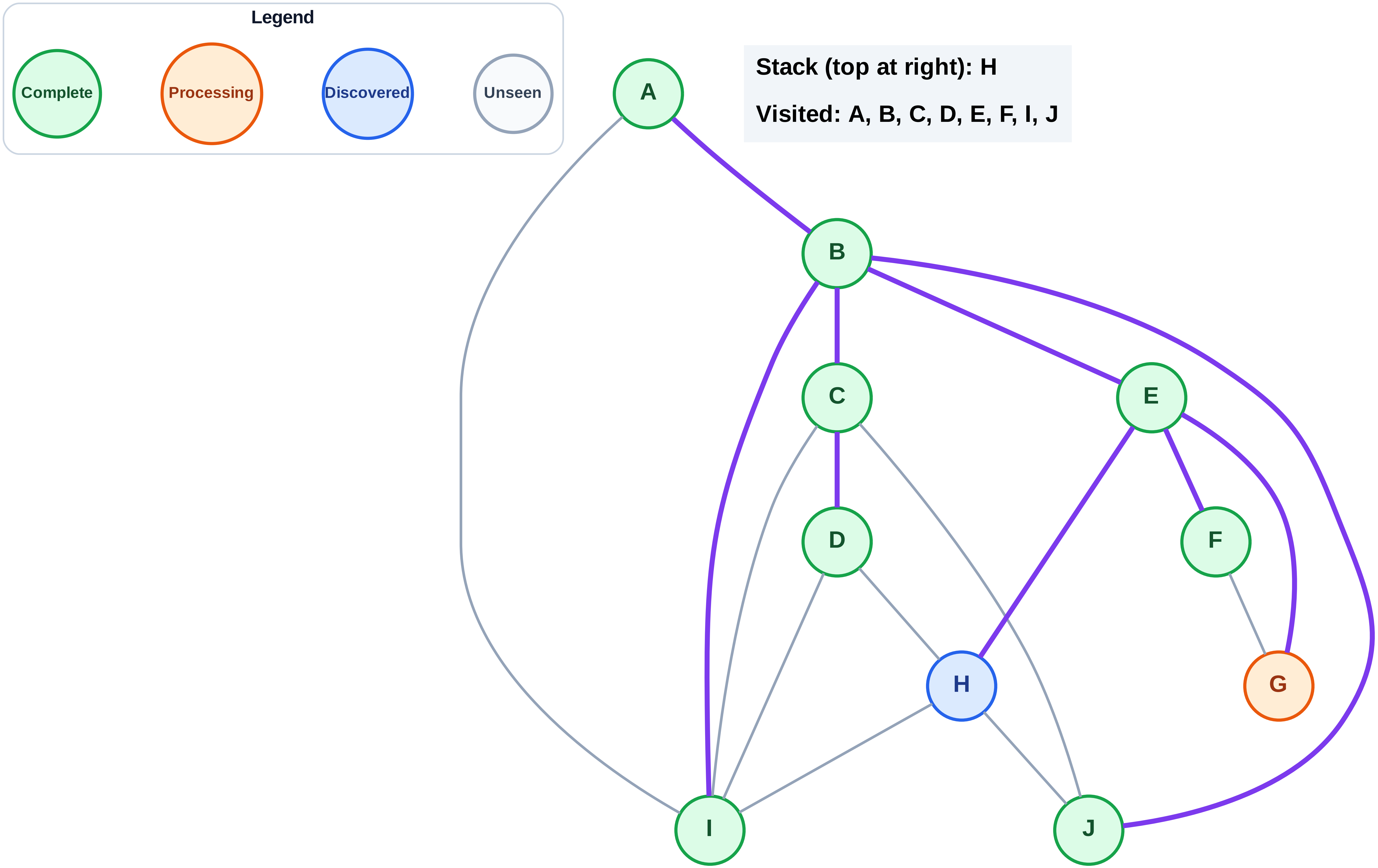
DFS Traversal

Step 17: No new neighbors from F



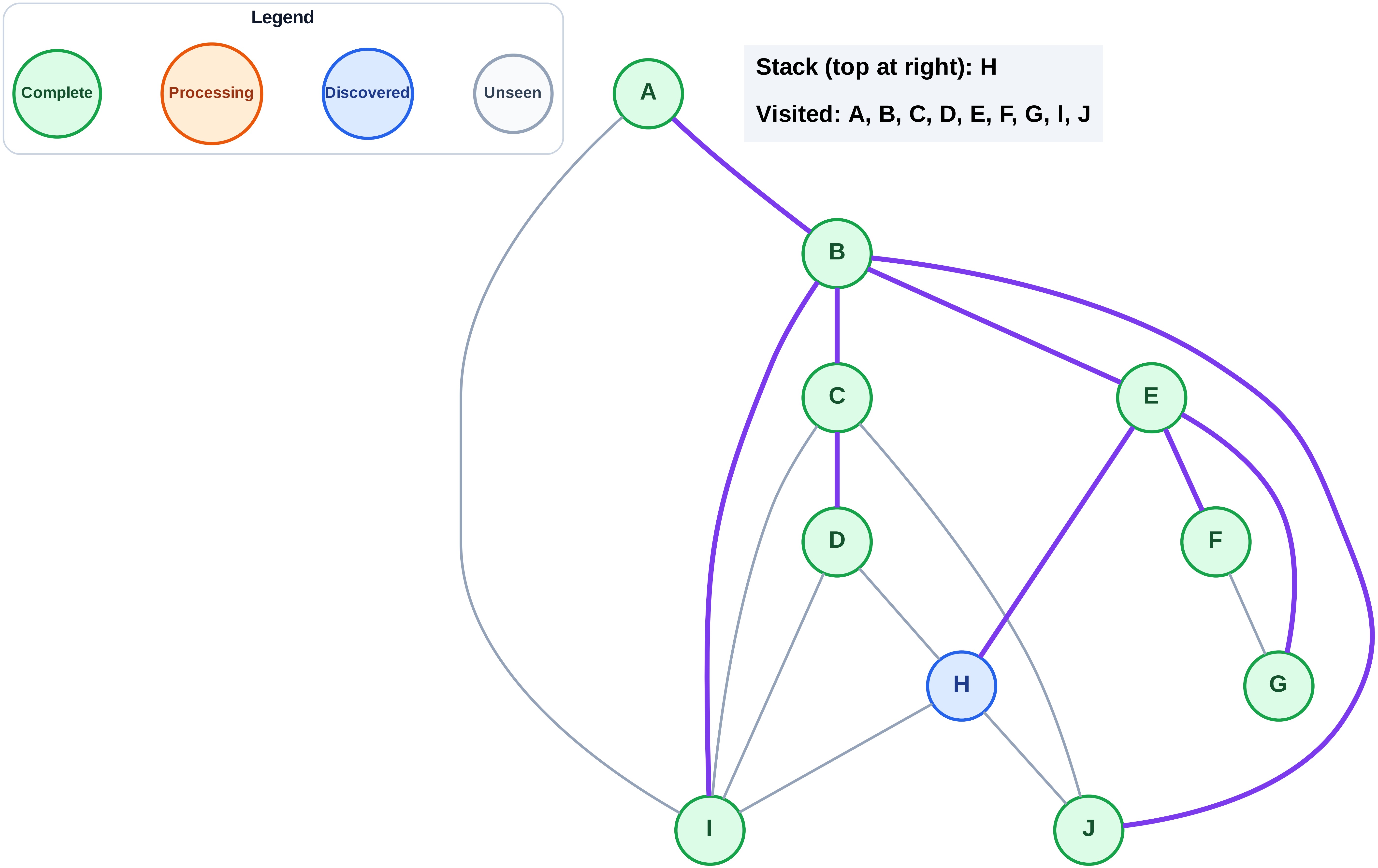
DFS Traversal

Step 18: Process node G



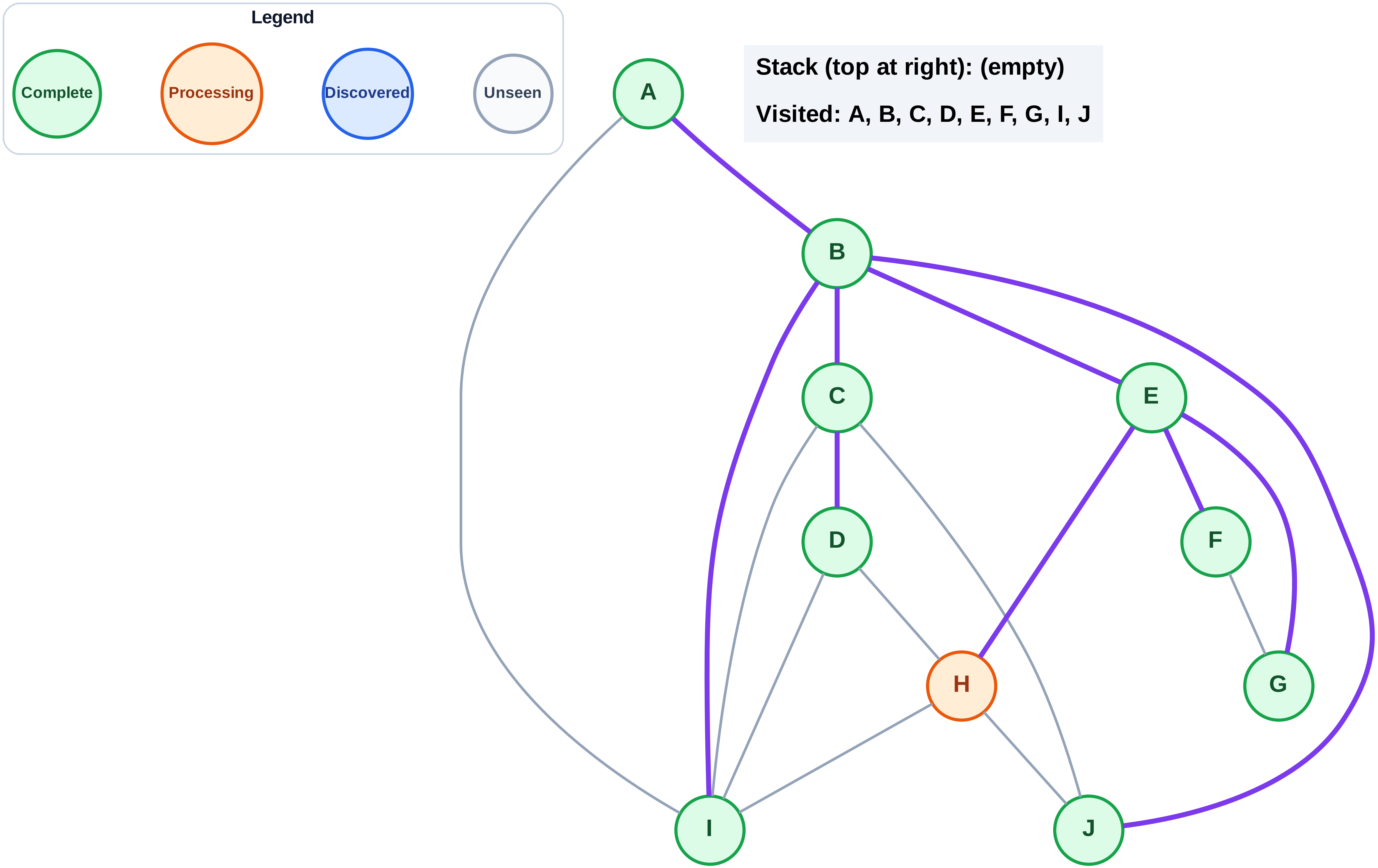
DFS Traversal

Step 19: No new neighbors from G



DFS Traversal

Step 20: Process node H



DFS Traversal

Step 21: No new neighbors from H